

Electricity network

Performance Report

2009 – 2010



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1. Profile

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1. Profile

Country Energy is a leading Australian energy services provider owned by the NSW Government.

With annual revenue of more than \$2 billion, Country Energy is one of the nation's largest utility businesses. Country Energy's success is driven by commercial decisions that meet local priorities and the needs of our customers, employees, stakeholders and shareholders.

Country Energy's core business is to operate a safe and reliable electricity, water and gas network service for customers.

Country Energy operates Australia's largest electricity distribution network, extending across 95 per cent of the NSW land mass and into parts of Queensland, Victoria and the Australian Capital Territory. This network is unique in Australia, with medium to very low customer density and wide variations in topographical and climatic conditions. It includes:

- More than 200,000 kilometres of powerlines, 97 per cent of which are overhead and three per cent underground
- Approximately 1.4 million power poles and more than 147,000 street lights
- Some 134,000 distribution substations and more than 300 zone substations. The majority of these substations are remotely monitored and controlled by Supervisory Control and Data Acquisition (SCADA) systems
- A diverse range of asset types, from single wire earth return (SWER) through to single and three phase, operating at voltage levels of 220kV, 132kV, 110kV, 66kV, 33kV, 22kV, 19.1kV, 12.7kV, 11kV, 6.6kV and 3.3kV
- Delivery of around 12,000 GWh of energy per year.

Some network components have an age of greater than 70 years, 50 per cent of overhead powerlines are more than 40 years old and the overall weighted average asset age is around 27 years.

Country Energy is investing almost \$4 billion between 2009 and 2014 to further strengthen and grow the network, almost doubling the previous five-year network investment of \$2.1 billion. In the process, the 20th century network is being transformed into a responsible Intelligent Network for the 21st century in order to deliver cost effective alternatives to traditional investment, improve reliability, support the growth of renewable energy and make energy efficiency simpler for customers.

Country Energy also provides reticulated natural gas to more than 25,000 customers in southern NSW and water and sewerage services to 10,000 customers in far western NSW.

Country Energy also provides competitive energy offers and value-added services – including renewable energy, LPG, energy efficiency advice, water management solutions and flexible payment options – to customers in six states and territories.

Country Energy continues to support a decentralised regional management structure to serve a customer base of approximately 800,000 customers. It's 4,500 employees live and work in local communities across rural and regional NSW, operating from 146 Customer and Field Service Centres, eight regional offices and several metropolitan locations to better respond to customer needs and expectations.



1. Profile

Table 1.1 - Network Operator Statistics

	Number at 30 6 09	Number at 30 6 10
Distribution Customer Numbers (Total)	786,241	801,913
Customer Numbers at Year End (North Western)	58,507	60,168
Customer Numbers at Year End (Southern)	111,230	115,308
Customer Numbers at Year End (Central Western)	91,205	94,144
Customer Numbers at Year End (Northern)	93,349	94,782
Customer Numbers at Year End (South Eastern)	119,941	120,081
Customer Numbers at Year End (Far North Coast)	136,943	139,294
Customer Numbers at Year End (Mid North Coast)	154,274	157,109
Customer Numbers at Year End (Far West)	20,792	21,026
Maximum Demand (MW)	2,332	2,239
Energy Received by Dist Network to Year End GWh	13,076	12,670
Energy Distributed to Year End (Residential) GWh	4,839	4,675
Energy Distributed to Year End (Non-Residential including un-metered supplies) GWh	7,285	7,041
Energy Distributed to Year End (North Western) GWh	1,172	1,147
Energy Distributed to Year End (Southern) GWh	1,857	1,868
Energy Distributed to Year End (Central Western) GWh	2,632	2,540
Energy Distributed to Year End (Northern) GWh	1,394	1,355
Energy Distributed to Year End (South Eastern) GWh	1,396	1,272
Energy Distributed to Year End (Far North Coast) GWh	1,522	1,489
Energy Distributed to Year End (Mid North Coast) GWh	1,673	1,607
Energy Distributed to Year End (Far West) GWh	478	438
System Loss Factor (%)	7.9	7.5
Transmission System (km)	–	–
Transmission Substation (Number) ¹	17	20
Sub Transmission System (km)	12,277	12,352
Substation - Zone (Number)	321	322
Substation - Distribution (Number)	132,212	134,524 ²
High Voltage Overhead (km)	144,546	144,661
High Voltage Underground (km)	1,795	1,911
Low Voltage Overhead (km) ³	26,308	26,939
Low Voltage Underground (km)	4,897	4,981
Poles (Number)	1,373,649	1,387,734 ⁴
Streetlights (Number)	143,753	147,733 ⁵
Employees (Full Time Equivalent Number)	4,278	4,474
Contractors (Full Time Equivalent Number)	267	454

¹ Country Energy assumes any substation that converts to a voltage that is not used for distribution is a Transmission Substation.

² Figure includes substations with a status of “out of service”. Substations with this status were not included in the figure reported as at 30 June 2009 and represent an increase of 2,100 substations.

³ Line length does not include LV services. Approx length of LV services is 14,000km.

⁴ Figure includes poles with a status of “out of service”. Poles with this status were not included in the figure reported as at 30 June 2009 and represent an increase of 12,000 poles.

⁵ Figure includes streetlights with a status of “out of service”. Streetlights with this status were not included in the figure reported as at 30 June 2009 and represent an increase of 1,200 streetlights.

2. Network Management

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countryenergy

2. Network Management

2.1 OVERVIEW

Country Energy manages a complex and geographically diverse power supply network. The distribution area is one of the largest in the world. To ensure decisions are localised and reflect the needs and concerns of the local population, the network is managed through a regional management structure.

Country Energy's Network Asset Management Plan has been put in place to meet its statutory and public responsibilities and to provide a safe, reliable and sustainable electricity supply to customers. The plan is divided into four chapters, these being:

- Network Safety and Reliability – aims to establish a framework that ensures Country Energy's sub-transmission and distribution system provides an adequate, reliable and safe supply of electricity of appropriate quality. The plan is revised regularly and management and operations employees are audited to ensure work practices are in accordance with the objectives of the plan (see sections 3 and 4).
- Customer Installation Safety Plan – ensures provision of safe electrical installations within Country Energy's network area and their safe connection to Country Energy's sub-transmission and distribution system to help provide a safe working environment for employees, service providers, contractors, customers and the general public (see section 6).
- Public Electrical Safety Awareness Plan – provides a framework and strategies to warn the public of the hazards associated with electricity and, in particular, the hazards associated with overhead powerlines, and to provide simple but effective ways to minimise their risk exposure (see section 9).
- Bushfire Risk Management Plan – aims to ensure that Country Energy's assets are managed in a way that will minimise the risk of bushfires, as well as protect its assets and maintain customer supply reliability at times of bushfire (see section 8).

Country Energy's commitment to safety management procedures can also be seen in its continuous improvements in safety measures (see section 5).

2.2 INDEPENDENT APPRAISAL PROCESS

Country Energy has engaged the services of Sinclair Knight Merz Pty Ltd ('SKM') to conduct an independent appraisal of this Network Performance Report 2009/2010. The appraisal has been conducted with due regard to the requirements of the Electricity Supply (Safety and Network Management) Regulation as well as the Independent Appraisal Guidelines given in the Electricity Network Performance Report Outline issued by Industry and Investment NSW.

2.3 NETWORK COMPLAINTS

Complaint information is taken directly from customers – either in person, over the phone, from the contact form on our website or from written complaints received. All complaints, including those resolved at first contact, are entered into Country Energy's dedicated Contact Management System (CMS). The complaint is then assigned to the most appropriate actioning officer, depending on the category of the complaint. An automatic escalation and reminder system ensures the complaint is dealt with in a timely manner.

All contact with the customer in the course of any investigation is recorded and contained within the CMS database.

Country Energy's system is designed to collect a variety of data and to store information in separate fields within the complaint database. This information can later be examined and sorted to provide the required statistics and trends and to help identify any systemic issues. It is also used as the basis for regulatory compliance reporting. Any trends or systemic issues identified are communicated, together with recommended actions, through a variety of reports.

The total network complaints trend continues to increase due to higher community expectations associated with reliability and quality of supply. Customer expectation associated with a more reliable system also can be attributed to rising energy costs.

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Complaints related to voltage issues appear to have stabilised with a marked reduction compared with the considerable increase in these complaints over the previous period.

The increase in complaints associated with network reliability appears to have been the result of a number of severe storm events that were experienced in this reporting period around the Mid and Far North Coast regions.

There was also a marked increase in complaints related to the necessary interruption to supply for planned works for the upgrade of the network. All complaints are individually addressed by each region and a customer's individual needs are taken into consideration. It is anticipated these types of complaints will continue as the delivery of the five-year plan necessitates further interruptions to supply.

Table 2.1 – Complaints received regarding network issues

	Previous years				Current year
	2005–2006	2006–2007	2007–2008	2008–2009	2009–2010
Complaints total	2,541	2,046	2,327	3,232	3,599
Complaints per 1,000 distribution customers	3.3	2.7	3.0	4.2	4.5
Complaints regarding vegetation management	105	152	164	169	234

Table 2.2 – Network complaint investigations completed 2009–2010

Category	Nature of complaint	Number	Number valid*
Voltage	Sustained over voltage	126	74
	Sustained under voltage	117	76
	Voltage fluctuations	293	134
	Voltage dips	77	45
	Switching transients	–	–
	N-E voltage difference	18	10
	Ground fault voltage	2	1
	Voltage unbalance	10	2
	Mains signalling voltages (outside defined range)	–	–
	HV injection (HV/LV intermix)	–	–
	Notching	–	–
	Invalid	301	–
	Sub total (supply voltage complaints)	643	342
Current	Direct current	–	–
	Harmonic content	1	–
	Inter harmonics	–	–
	Invalid	1	–
	Sub total (supply current complaints)	1	–

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Table 2.2 – Network complaint investigations completed 2009–2010

Category	Nature of complaint	Number	Number valid*
Other Quality	Mains signalling reliability	9	9
	Noise and Interference	121	49
	Level of supply capacity	53	24
	Supply frequency	–	–
	Level of EMF	1	–
	Invalid	102	–
	Sub total (other quality of supply complaints)	184	82
Sub total (all quality of supply complaints)		827	424
Reliability	No. of supply failures	234	130
	Duration of supply failures	45	23
	No. of <1 min. interruptions	181	110
	Invalid	197	–
Sub total (reliability of supply)		460	263
Safety	Overhead line safety	–	–
	Underground safety	2	2
	Electrical station safety	–	–
	Service line safety	–	–
	Invalid	–	–
Sub total (network safety)		2	2
Total		1,290	689

* A complaint is valid where non-compliance with published service and network standards occur

2.4 CUSTOMER SERVICE STANDARDS REPORTING

Table 2.3 – Customer Service Standards 2009/2010 Data

	Payments given based on interruption duration (total number)	Payments given based on interruption frequency (total number)	Claims not paid based on interruption duration (total number)	Claims not paid based on interruption frequency (total number)
Metropolitan	N/A	N/A	N/A	N/A
Non-Metropolitan	11	–	13	4

Table 2.3 shows that there were a total of four claims associated with interruption frequency that were denied because of insufficient outages recorded. In addition, one of the claims was submitted after the three-month cut-off date (30 September 2009) following the end of the financial year in which the outages occurred.

Of the 13 duration claims that were denied 12 were the result of severe weather events, while the remaining claim was declined on the basis of there not being any recorded outages greater than 18 hours in duration.

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3. Network Planning

3.1 OVERVIEW

Country Energy is committed to providing a safe, secure and reliable supply of energy in a cost-effective manner. Country Energy achieves this by planning its network and making investment decisions in line with its *CEOP8018 Network Asset Management Plan (NAMP)*.

The *NAMP* outlines Country Energy's obligations in relation to network planning, including the need to compile and publish an annual *Electricity System Development Review (ESDR)*. The *NAMP* has been prepared in accordance with the compliance obligations of network management legislation, regulations and related codes of practice, and reflects recognised industry best practices and standards relating to the management of electricity infrastructure assets. It is designed to conform with NSW Government policy and planning guidelines on total asset management, which addresses strategic planning relating to capital investments, renewal and maintenance.

Country Energy's network development is undertaken in accordance with the *Electricity Supply Act*, the *National Electricity Code*, the *NSW Code of Practice – Demand Management for Electricity Distributors* and policy *CEPG8003 Sub-transmission and Distribution Network Planning Criteria and Guidelines*.

In general, Country Energy plans the development of its network to ensure:

- network capacity is adequate to meet power transfer requirements
- electrical and thermal design ratings (normal and overload) of equipment are not exceeded
- supply reliability is in accordance with published standards, or as negotiated to meet the special requirements of individual major network customers
- quality of supply meets published standards and system voltage levels are maintained within acceptable standard limits
- safety standards are maintained or exceeded
- environmental constraints are satisfied
- the above requirements are met in a cost effective manner.

3.2 DESIGN PLANNING CRITERIA

In August 2005, the Minister for Energy and Utilities introduced new *licence conditions* that included requirements for planning and design of sub-transmission and distribution networks.

Following a review of the *licence conditions* conducted by the Minister, the conditions were replaced with updated and revised conditions with effect from 1 December 2007.

The *design planning criteria* set out:

- input standards to be used by a licence holder in planning its network, and
- requirements for load forecasting and contingency planning methodologies intended to achieve operational outcomes.

The baseline levels of planned redundancy required under the design planning criteria underpin Country Energy's plans for the network to ensure, as far as is reasonably practicable, that:

- the *reliability standards* are met
- an adequate supply, with an appropriate level of redundancy consistent with its regulatory obligations, is provided.

The *design planning criteria* applicable to Country Energy are listed in Table 3.1.

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Table 3.1 – Design planning criteria

Network element	Load type	Forecast demand or expected demand	Security standard	Customer interruption time
Sub transmission line	Urban and non-urban	≥ 15 MVA	N-1 ¹	< one minute
	Urban and non-urban	< 15 MVA	N ²	<i>Best practice repair time</i>
Sub Transmission Substation	Urban and non-urban	Any	N-1	< one minute
Zone Substation	Urban and non-urban	≥ 15 MVA	N-1 ¹	< one minute
	Urban and non-urban	< 15 MVA	N ²	<i>Best practice repair time</i>
Distribution Feeder	Urban (regional centres) ⁴	Any	N-1 ³	< four hours
	Urban (other) and non-urban	Any	N	<i>Best practice repair time</i>
Distribution Substation	Urban and non-urban	Any	N	<i>Best practice repair time</i>

1. For a *sub-transmission line - overhead* and a zone substation:

- Under N-1 conditions, the *forecast demand* is not to exceed the *thermal capacity* for more than one per cent of the time i.e. a total aggregate time of 88 hours per annum, up to a maximum of 20 per cent above the *thermal capacity* under N-1 conditions. For Country Energy, in other than regional centres, the *forecast demand* must not exceed the *thermal capacity* under N-1 conditions.
- Under N conditions, a further criterion is that the *thermal capacity* is required to meet at least 115 per cent of *forecast demand*.

For a *sub-transmission line – underground*, any overhead section may be designed as if it was a *sub-transmission line – overhead*, providing the *forecast demand* does not exceed the *thermal capacity* of the underground section at any time under N-1 conditions.

- Under N conditions, *thermal capacity* is to be provided for greater than 115 per cent of *forecast demand*.
- By 30 June 2014, expected demand is to be no more than 80 per cent of feeder *thermal capacity* (under system normal operating conditions) with switchable interconnection to adjacent feeders enabling restoration for an unplanned network element failure. By 30 June 2019, expected demand is to be no more than 75 per cent of feeder *thermal capacity*. In order to achieve compliance, feeder reinforcement projects may need to be undertaken over more than one *regulatory period*. In those cases where a number of feeders form an interrelated system (such as a meshed network), the limits apply to the average loading of the feeders within the one system.
- Regional centre means: Albury, Armidale, Ballina, Bathurst, Broken Hill, Coffs Harbour (including Sawtell), Dubbo, Forster-Tuncurry, Goulburn, Grafton, Griffith, Lismore, Orange, Port Macquarie, Queanbeyan, Tamworth, Taree, Tweed Heads and Wagga Wagga.

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3.2.1 Design planning criteria compliance reporting

sub-transmission planning

The planning criterion considers the adequacy of the sub-transmission network not only to meet the energy demand requirement, but also its capability to do so within component electrical and thermal ratings and voltage limits. Analysis is carried out to generally achieve an economically efficient outcome in which the sub-transmission network is secure and within rating following the forced outage of any single circuit line or substation element during peak periods.

Provision is also made for use of the short-time capability or cyclic rating of transformers during a planned outage of a parallel transformer.

Other sub-transmission planning criteria outline that:

- network equipment is designed to withstand maximum fault duty
- any sub-transmission fault is seen by at least two protection systems that are fully independent.

Each potential augmentation project is treated on its own merits. The main economic impact of sub-transmission investment is on a reduction in the value of the energy not supplied (the 'unserved energy'), an extension of the time taken for capacity limits to be exceeded and the initial capital costs. These are quantified over the expected life of the project and costed at appropriate rates.

Planning of the sub-transmission network is usually carried out over a longer time horizon of between five to ten years. Country Energy distinguishes between long-term and short-term network planning, with long-term network planning generally confined to 132kV and 66kV networks. Short-term network planning is usually limited to networks ranging from 11 to 33kV, with a horizon period of up to five years.

Primary distribution planning

The planning and capital investment framework associated with Country Energy's distribution network is principally driven by technical considerations and requirements. This is primarily due to the inherent geographical challenges, our commitment to providing customers with published standards of reliability and quality of supply, and low profit structure of the network that is essentially radial.

Distribution network planning is generally of a short-term nature – up to five years. Country Energy's traditional approach to distribution network planning has been to focus on feeder augmentation works to provide supply to new or growing loads, maintain adequate voltage regulation, enhance reliability of supply, upgrade *thermal capacity* (and minimise system losses). This form of distribution planning criteria will continue.

Network elements not complying with the design planning criteria on 1 July 2009

For any new network element where planning commenced after 1 December 2007, Country Energy is required to comply with the design planning criteria. Plans have been developed to ensure that existing elements detailed in Tables 3.2 to 3.4 will be compliant by 1 July 2014 when the *licence conditions* apply to these elements.

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Table 3.2 – Sub-transmission lines and substations and zone substations

Each network element or class of network elements

not complying with the design planning criteria on 1 July of the current year (2009/2010)

Element including location, customer numbers, element length/capacity	Description of non-compliance and reason	Proposed remedial actions and timetable
Forbes – West Jemalong 66kV feeder (37km) 7,000 customers Central West Region	N-1 ≥ 15MVA sub-transmission feeder. Radial 66kV feeder with peak demand of 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Forbes – West Jemalong 66kV feeder, expected completion 2011/12
TransGrid–Orange South–Orange West 66kV feeders (20km) 12,000 customers Central West region	N-1 ≥ 15MVA sub-transmission feeder. Orange South and Orange West both have peak demands above 15MVA, 66kV feeders cannot be restored within one minute during contingency	Augment Orange South and Orange West zone substation, add 66kV feeder bays, expected completion 2012/13
Bathurst Russell St zone substation 8,000 customers Central West region	N-1 ≥ 15MVA zone substation. Russell St zone substation has a firm transformer capacity of 20MVA with a peak demand of 26MVA	Augment Russell St zone substation to 30MVA capacity, transfer load to adjacent zone substation, expected completion 2011/12
TransGrid – Parkes 66kV feeder (12km) 7,000 customers Central West region	N-1 ≥ 15MVA sub-transmission feeder. 66kV feeder has peak demand of 20MVA. Outage of 66kV feeder results in overload of remaining in-service 66kV feeders	Construct a second 66kV feeder to Parkes town, expected completion 2011/12
Temora – Thanowring 66kV feeder (8km) 7,000 customers Central West region	N-1 ≥ 15MVA sub-transmission feeder. Radial 66kV feeder with peak demand of 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Temora – Thanowring 66kV feeder, expected completion 2012/13
Koolkhan – Maclean 66kV feeder (36km) 11,000 customers Far North Coast region	N-1 ≥ 15MVA sub-transmission feeder. Radial 66kV feeder with peak demand above 20MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Koolkhan – MacLean 66kV feeder, expected completion 2010/11
Lismore 132kV – Lismore South 66kV feeders (2x3km) 17,000 customers Far North Coast region	N-1 ≥ 15MVA sub-transmission feeder. 66kV network to Lismore South has peak demand of 35+MVA. Outage of 66kV feeder results in overload of remaining in-service 66kV feeder	Augment Lismore 132kV – Lismore South 66kV feeders, expected completion 2010/11
Ewingsdale – Mullumbimby 66kV feeder (13km) 20,000 customers Far North Coast region	N-1 ≥ 15MVA sub-transmission feeder. Outage of 66kV feeder results in voltage levels below standard and overload of remaining in-service 66kV feeders	Staged conversion of 66kV network to 132kV, expected completion of stage one in 2012/13

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Table 3.2 – Sub-transmission lines and substations and zone substations

Each network element or class of network elements

not complying with the design planning criteria on 1 July of the current year (2009/2010)

Element including location, customer numbers, element length/capacity	Description of non-compliance and reason	Proposed remedial actions and timetable
Merbein – Dareton 66kV feeder (Buronga load) Far West region	N-1 ≥ 15MVA sub transmission feeder. 66kV radial feeder with a peak demand of 30+MVA. Outage of 66kV feeder leads to loss of supply to all customers	Establish Buronga 66/22kV zone substation, construct 66kV feeder from Gol Gol to Buronga, reconfigure 22kV feeder to 66kV, expected completion 2010/11
Karangī – Coffs Harbour North 66kV feeder (11km) 10,000 customers Mid North Coast region	N-1 ≥ 15MVA sub transmission feeder. 66kV feeder has peak demand of 30MVA. Outage of 66kV feeder results in overload of remaining in-service 66kV feeders and extreme low voltage levels	Construct a Coffs Harbour North – Coffs Harbour South 66kV feeder, expected completion 2011/12
Stroud – Gloucester 33kV feeders (2 x 40km) 4,000 customers Mid North Coast region	N-1 ≥ 15MVA sub-transmission feeder. 33kV feeders have peak demand of 15MVA. Outage of 33kV feeder results in voltage levels below standard and overload of remaining in-service 33kV feeder	Establish a 33kV switching station at Wards River, expected completion 2010/11
Taree – Kew 66kV feeder (45km) 7,000 customers Mid North Coast region	N-1 ≥ 15MVA sub-transmission feeder. 66kV feeder has peak demand of 15MVA. Outage of 66kV feeder results in voltage levels below standard and overload of remaining in-service 66kV feeder	Establish a 132/66/11kV transmission substation at Herons Creek, expected completion 2013/14
Taree – Failford - Tuncurry – Forster – Bohnock 66kV feeders (100km) 18,000 customers Mid North Coast region	N-1 ≥ 15MVA sub-transmission feeder. 66kV feeders have peak demand of 40+MVA. Outage of 66kV feeder results in voltage levels below standard and overload of remaining in-service 66kV feeders	Establish a 132/66kV transmission substation at Nahiack and construct interconnecting 66kV feeders, expected completion 2011/12
Forster – Tuncurry 66kV feeder (8km) 9,000 customers Mid North Coast region	N-1 ≥ 15MVA sub-transmission feeder. 66kV feeders have peak demand of 20MVA. Outage of 66kV feeder results in overload of remaining in-service 66kV feeder	Replace existing 66kV feeder river crossing with 66kV cable on bridge, expected completion 2010/11
TransGrid – Glen Innes 66kV feeder (9km) 7,000 customers Northern region	N-1 ≥ 15MVA sub-transmission feeder. 66kV feeder has peak demand of 15MVA. Limited capacity of remaining in-service 66kV feeder	Construct a second Trans Grid – Glen Innes 66kV feeder in joint planning with Trans Grid 132kV constraints, expected completion 2010/11
TransGrid – Gunnedah 66kV feeders (2 x 12km) 7,000 customers Northern region	N-1 ≥ 15MVA sub-transmission feeder. 66kV feeders have peak demand of 23MVA. Outage of 66kV feeder results in overload of remaining in-service 66kV feeder	Upgrade Trans Grid – Gunnedah 66kV feeders, expected completion 2010/11

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Table 3.2 – Sub-transmission lines and substations and zone substations

Each network element or class of network elements

not complying with the design planning criteria on 1 July of the current year (2009/2010)

Element including location, customer numbers, element length/capacity	Description of non-compliance and reason	Proposed remedial actions and timetable
Borthwick St – Wynne St 66kV feeder, Wynne St zone substation 8,000 customers Northern region	N-1 $\geq$ 15MVA sub-transmission feeder and N-1 $\geq$ 15MVA zone substation. Wynne St substation with a peak demand of 25MVA is controlled by single 66kV circuit breaker at Borthwick St. Transformer or feeder fault leads to loss of supply to all customers with restoration greater than one minute	Augment Wynne St and Borthwick St substations and loop in/out Ashford 66kV feeder to Wynne St, expected completion 2012/13
TransGrid – South Tamworth – Oxley Vale 66kV feeders (25km) 11,000 customers Northern region	N-1 $\geq$ 15MVA sub transmission feeder. 66kV feeders have peak demand of 40MVA. Outage of 66kV feeder results in overload of remaining in-service 66kV feeders and supply cannot be restored in one minute	Augment Trans Grid – South Tamworth – Oxley Vale feeders and configure substation to 66kV 'ring' network, expected completion 2011/12
TransGrid – Goddard Lane 66kV feeder (10km) 5,000 customers Northern region	N-1 $\geq$ 15MVA sub-transmission feeder. Radial 66kV feeder with peak demand above 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Trans Grid – Goddard Lane 66kV feeder, expected completion 2011/12
TransGrid – Quirindi 66kV feeder (56km) 7,000 customers Northern region	N-1 $\geq$ 15MVA sub-transmission feeder. Radial 66kV feeder with peak demand of 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Trans Grid – Quirindi 66kV feeder, expected completion 2012/13
Beryl – Mudgee (tee) 132kV feeder (35km) 9,000 customers North West region	N-1 $\geq$ 15MVA sub-transmission feeder. 132kV feeder has peak demand of 25MVA. Outage of 132kV feeder results in overload of remaining in-service 66kV feeder	Augment Beryl – Mudgee sub transmission network, expected completion 2012/13
Beryl – Dunedoo 66kV feeder (41km) 7,000 customers North West region	N-1 $\geq$ 15MVA sub-transmission feeder. Radial 66kV feeder with peak demand above 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Beryl – Dunedoo 66kV feeder, expected completion 2012/13
Nyngan – Cobar 66kV feeder (130km) 4,000 customers North West region	N sub-transmission feeder. Radial 66kV feeder with voltage levels below standard during system normal operation	Install a 66kV regulator on Nyngan – Cobar 66kV feeder, expected completion 2011/12
Dubbo – Yarrandale 66kV feeder (5.5km) 9,000 customers North West region	N-1 $\geq$ 15MVA sub-transmission feeder. Radial 66kV feeder with peak demand above 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Dubbo – Yarrandale 66kV feeder, expected completion 2010/11

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Table 3.2 – Sub-transmission lines and substations and zone substations
Each network element or class of network elements
not complying with the design planning criteria on 1 July of the current year (2009/2010)

Element including location, customer numbers, element length/capacity	Description of non-compliance and reason	Proposed remedial actions and timetable
Yarrandale – Gilgandra 66kV feeder (65km) 5,000 customers North West region	N-1 $\geq$ 15MVA sub-transmission feeder. Radial 66kV feeder with peak demand above 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Yarrandale – Gilgandra 66kV feeder, expected completion 2012/13
Wellington – Dubbo 132kV feeders (2 x 47km) 70,000 customers North West region	N-1 $\geq$ 15MVA transmission feeder. 132kV feeder with peak demand above 150MVA, outage of a 132kV feeder leads to overload of remaining in-service 132kV feeder	Construct a Wellington – Narromine 132kV feeder, expected completion 2013/14
Narromine – Dubbo West sub transmission network 6,000 customers North West region	N-1 $\geq$ 15MVA zone substation. Dubbo West approaching 15MVA, currently supplied via a radial from Dubbo 132 to West Dubbo and to Narromine, provides N-1 by a 66kV supply at Narromine, via the establishment of a 132/66kV transformer. A switched 132kV busbar will provide a connection point for a future Wellington – Narromine 132kV feeder	Establish a 132/66kV transmission substation at Narromine 132kV tee, expected completion 2010/11
Yarrandale zone substation 4,000 customers North West region	Yarrandale has a single 10MVA transformer with peak demand approaching 10MVA, and expected additional load, a larger transformer and another transformer is required. A new second 66kV feeder from Dubbo 132kV to Yarrandale requires 66kV feeder bays at Yarrandale	Augment Yarrandale zone substation, expected completion 2010/11
Bega – Pambula – Eden South 66kV feeder (60km) South Eastern region	N-1 $\geq$ 15MVA sub transmission feeder. 66kV radial feeder with a peak demand of 20+MVA. Outage of 66kV feeder leads to loss of supply to all customers	Construct second Bega – Eden South 66kV feeder over redundant 33kV easement, expected completion 2011/12
Cooma – Bega 132kV feeder (92km) and Cooma – Bombala – Bega 66kV feeder (96km) 23,000 customers South East region	N-1 $\geq$ 15MVA sub transmission feeders. 132kV and 66kV network with peak demand of 50MVA. Outage of 132kV feeder leads to overload of 66kV feeder, results in loss load of 30+MVA	Construct a Cooma – Bega dual 132/66kV feeder on existing 66kV easement, expected completion 2011/12
Evans Lane – Batemans Bay 132kV feeder (57km) 25,000 customers South Eastern region	N-1 $\geq$ 15MVA sub transmission feeder. 132kV network with peak demand of 50+MVA. Outage of 132kV feeder leads to extreme low voltage levels and possible voltage collapse	Rebuild smaller 132kV feeder with larger conductor and staged power factor correction, expected completion 2012/13. (Project may change dependent on projects in Integral Energy 132kV network)

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Table 3.2 – Sub-transmission lines and substations and zone substations

Each network element or class of network elements

not complying with the design planning criteria on 1 July of the current year (2009/2010)

Element including location, customer numbers, element length/capacity	Description of non-compliance and reason	Proposed remedial actions and timetable
Mulwala – Finley 132kV and 66kV feeders (60km) 7,000 customers Southern region	N-1 ≥ 15MVA sub transmission feeders. Both Finley zone substation and Mulwala 132/66kV substation have demands above 15MVA. On loss of the 132kV feeder to Mulwala and the 66kV feeder to Finley, the remaining in-service feeder is overloaded	Establish a 132/22kV substation at Mulwala, construct a 132kV feeder from Mulwala tee to new 132/22kV substation, construct 132/66kV dual feeder from Finley 132kV to Finley zone substation, convert 66kv Finley – Mulwala feeder to 132kV, expected completion 2011/12
Griffith 33kV feeders (37km) 12,000 customers Southern region	N-1 ≥ 15MVA sub transmission feeders. 33kV feeders have peak demand above 15MVA. Outage of 33kV feeder results in overload of remaining in-service 33kV feeders	Initially construct 132kV feeder operated at 33kV. Establish a Griffith West 132/33kV transmission substation and 33kV feeder interconnections into the long term, expected initial feeder completion 2012/13
Deniliquin to Moama 66kV feeder (70km) 4,000 customers Southern region	N-1 ≥ 15MVA sub transmission feeder. 66kV radial feeder with a peak demand of 15+MVA. Outage of 66kV feeder leads to loss of supply to all customers	Construct a second Deniliquin to Moama 66kV feeder, expected completion 2011/12
Ashmont zone substation 15,000 customers (Wagga City) Southern region	N-1 ≥ 15MVA sub transmission feeders. The Wagga Wagga city 66kV network is arranged in a ‘tee’ step ladder formation, with some zone substations not having incoming 66kV feeder or 66kV transformer protection. Up to four zone substations can be affected by a single feeder or transformer fault. Restoration of supply within one minute is not possible	Augment Ashmont substation to fully switched 66kV substation and construct 66kV feeder connection, expected completion 2010/11
Wagga – Temora 132kV feeder (80km) 9,000 customers Southern region	N-1 ≥ 15MVA sub transmission feeder. 132kV feeder with limited 66kV backup, Peak demand of 50MVA. Outage of 132kV feeder leads to overload of 66kV feeder and results in loss load of 20+MVA	Rebuild existing Wagga Wagga – Junee – Temora 66kV feeder as 132/66kV dual circuit, expected completion 2012/13
Wagga – Lockhart – Kywong – Narrandera – Yanco 66kV feeder (85km) 5,000 customers Southern region	N-1 ≥ 15MVA sub transmission feeder. 66kV backup to Narrandera limited by low voltage levels	Install a 66kV regulator on Wagga Wagga – Narrandera 66kV feeder near Lockhart, expected completion 2010/11

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Table 3.2 – Sub-transmission lines and substations and zone substations

Each network element or class of network elements

not complying with the design planning criteria on 1 July of the current year (2009/2010)

Element including location, customer numbers, element length/capacity	Description of non-compliance and reason	Proposed remedial actions and timetable
Bourkelands zone substation 15,000 customers (Wagga Wagga city) Southern region	N-1 $\geq$ 15MVA sub transmission feeders. The Wagga Wagga city 66kV network is arranged in a 'tee' step ladder formation, with some zone substations not having incoming 66kV feeder or 66kV transformer protection. Up to four zone substations can be affected by a single feeder or transformer fault. Restoration of supply within one minute is not possible	Augment Bourkelands substation with fully switched in/out 66kV busbar and construct Ashmont to Bourkelands 66kV line - expected completion 2010/11
Bomen zone substation 15,000 customers (Wagga Wagga city) Southern region	N-1 $\geq$ 15MVA sub transmission feeders. The Wagga Wagga city 66kV network is arranged in a 'tee' step ladder formation, with some zone substations not having incoming 66kV feeder or 66kV transformer protection. Up to four zone substations can be affected by a single feeder or transformer fault. Restoration of supply within one minute is not possible	Augment Bomen substation, expected completion 2010/11
Cartwrights Hill zone substation 15,000 customers (Wagga Wagga city) Southern region	N-1 $\geq$ 15MVA sub transmission feeders. The Wagga Wagga city 66kV network is arranged in a 'tee' step ladder formation, with some zone substations not having incoming 66kV feeder or 66kV transformer protection. Up to four zone substations can be affected by a single feeder or transformer fault. Restoration of supply within one minute is not possible	Augment Cartwrights Hill substation with fully switched in/out 66kV busbar, expected completion 2011/12
Wagga Wagga – Uranquinty 66kV feeder (19km) 7,000 customers Southern region	N-1 $\geq$ 15MVA sub transmission feeder. Radial 66kV feeder with peak demand of 18MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct Bourkelands – Uranquinty 66kV feeder and augment Uranquinty 66kV busbar, expected completion 2011/12
Wagga Wagga city network 66kV feeders (5km) 15,000 customers (Wagga Wagga city) Southern region	N-1 $\geq$ 15MVA sub transmission feeders. The Wagga Wagga city 66kV network is arranged in a 'tee' step ladder formation, with some zone substations not having incoming 66kV feeder or 66kV transformer protection. Up to four zone substations can be affected by a single feeder or transformer fault. Restoration of supply within one minute is not possible	Construct a number of short 66kV feeders between zone substations at Ashmont, Bomen and Cartwrights Hill, expected completion 2011/12

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Table 3.3: Distribution feeder summary report by class of network elements

Not Complying with the Design Planning Criteria on 1 July of the Current Year (2009/2010)

Total number of feeders	Number of feeders without N-1 capability (1 minute)	Description and reason for non-compliance	Proposed remedial actions and timetable
N-1 Regional centres			
362	175	Insufficient feeder thermal ratings, lack of interconnectors, underrated interconnecting switchgear	Upgraded and additional feeders, construction of interconnectors, upgrading and additional interconnecting switchgear (gas switches, reclosers) N-1 programs have been set and are scheduled to be completed by 2014. There are 753 programs of work scheduled, of which 30 per cent have been completed
Total number of feeders	Number of feeders without N-1 capability (1 minute)	Description and reason for non-compliance	Proposed remedial actions and timetable
Urban⁶			
As above	As above	As above	As above
Non-urban			
1,359	66	Insufficient capacity due to incremental growth	As N capability is exceeded, remedial plans will be developed accordingly. All feeders without N Capability have augmentation works programmed

⁶By definition, urban for Country Energy means areas within a regional centre (page eight of the *licence conditions* 2007).

Table 3.4: Distribution Substation Summary Report by Class of Network Elements

Not Complying with the Design Planning Criteria on 1 July of the Current Year (2009/2010)

Total number of feeders	Number of feeders without N-1 capability (1 minute)	Description and reason for non-compliance	Proposed remedial actions and timetable
CBD			
Not applicable	Not applicable	Not applicable	Not applicable
Total number of feeders	Number of feeders without N-1 capability (1 minute)	Description and reason for non-compliance	Proposed remedial actions and timetable
Urban and non-urban			
132,212	246	Incremental lifestyle growth	Substations programmed to be upgraded when identified

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3.3 DEMAND MANAGEMENT

Where significant capital investment in Country Energy's sub-transmission and distribution systems is proposed to extend or increase supply capacity, consideration is given to whether it would be reasonable to expect that a non-network option could cost effectively avoid or postpone the project.

In accordance with the requirements of the *National Electricity Rules* and the *NSW Code of Practice, Demand Management for Electricity Distributors*, appropriate reports are published on the Country Energy website. The website also provides for non-network service providers to apply for inclusion on Country Energy's Register of Interested Parties and all registrants are informed of the report's availability.

Country Energy consults widely with non-network service providers to maintain currency with available technologies and their likely cost in order to be able to adequately assess cost effectiveness relative to a conventional network investment solution.

Initiatives that can be applied to address specific network constraints or customer groups include:

- Embedded (distributed) generation
- Emerging technologies
- Negotiated outcomes.

There are comparatively few large customers or embedded generators in Country Energy's network which limits opportunities for large scale demand reductions in locations that are subject to network constraints.

While there has been a significant increase in embedded generation through various renewable energy programs, the intermittent nature of their generation makes both photovoltaic and wind generation generally too unreliable for network support without the addition of significant energy storage. There is the prospect for solar thermal systems having inherent energy storage capability but these are still in development stage

Much of the capital investment in Country Energy's network is driven by N-1 reliability issues on radial sub-transmission lines where loading exceeds 15 MVA. This requires a demand management solution that can support the supplied load at all times or be fully deployed within 60 seconds (i.e. the equivalent of a second line with automatic changeover) and effectively limits the non network options to embedded base load generation, large scale energy storage or fast start up generation with short term support from energy storage. The market has not been able to provide a reliable, economic alternative for this class of constraint.

Another significant factor in capital investment within Country Energy is the replacement of older equipment for safety and operational reasons. In these types of projects where augmentation of equipment is a secondary component achieved at incremental cost there is little opportunity to source a timely or cost effective non network alternative.

Demand growth in much of Country Energy's network is the result of general population growth and also an increase in the average peak demand per connection as a direct result of lifestyle factors. It is estimated that an increase in peak demand of 1 kVA will, on average, require an investment of some \$2,500 in network augmentation to maintain adequate supply quality and meet regulated reliability standards. This figure will be higher or lower, depending on where the peak demand increase occurs on the network.

In particular, air conditioning has been a major contributor to an emerging summer peaking trend. This has proven to be difficult to address using conventional non-network options although advances in control flexibility and chiller technology may make this more viable in the future.

There are significant opportunities for Demand Management options which can be applied at a broader level to slow demand growth and address emerging distribution network constraints. This will require investment in the education of customers and the development of new, innovative technologies to support energy efficiency and small scale embedded generation initiatives. The Demand Management Innovation Allowance Scheme for the 2009/2014 regulatory period will be used to broaden our knowledge base in these areas in order to develop and promote non network alternatives for the longer term benefit of customers.

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Increase in renewable energy connection

Country Energy recognises that distributed energy resources, particularly renewables, will become increasingly important in reducing power flows and losses in the distribution networks. In order to minimise barriers to increased penetration of distributed energy resources, metering equipment for residential customers installing photovoltaic arrays is provided free of charge.

During 2009/2010, 12,600 applications for PV installations were approved with a total connected capacity of 35,900kW. While it is difficult to assess the long-term impact on network demands, it is anticipated that there will be some benefit in summer peaking areas. Our initial estimate is that network demand could be reduced by as much as 20 per cent of the connected PV capacity in locations where there is good overlap between generation and network loading.

Higher levels of embedded generation can create issues with voltage rise during periods of light loading, particularly in rural areas where distribution feeders, transformers and low voltage mains do not have significant capacity to transmit reverse power flows. This is exacerbated by the increasing number of applications for the connection of larger PV installations, between 5kW and 10kW, in more remote areas. Country Energy is investigating the potential for reactive power support from a four quadrant inverter as an alternative to network augmentation to mitigate voltage rises from embedded generation.

Demand management process

Country Energy has implemented internal procedures to comply with the Demand Management Code of Practice and *National Electricity Rules*. This ensures the network planning process is transparent, consultative and equitable in the treatment of demand management solutions relative to traditional network alternatives. This process includes:

- Publication of the Annual Electricity System Development Review
- Maintenance of a Register of Interested Parties
- Review of emerging constraints with an expected network solution in excess of \$1 million
- Publication of Consultation Papers via AEMO and Country Energy external web page
- Notification to Interested Parties of demand management opportunities
- Consultation with prospective demand management service providers.

Demand management initiatives

Country Energy is pro-active in seeking demand management solutions through participation in groups, the provision of policy advice and monitoring of emerging and innovative technologies. These groups include

- ENA Demand Management and Embedded Generation Committee (DMEGC)
- CSIRO working groups on demand management
- Collaborative research arrangements with leading academic institutions.

During 2009/2010, Country Energy sought expressions of interest for the provision of demand management services. Commercial arrangements are currently being negotiated with a number of companies that will be able to collectively provide expertise and advice on all aspects of non-network applications.

Inverter research and development

In 2008/2009, Country Energy formed an alliance with an Australian power electronics designer and manufacturer for the development of a grid interactive inverter to provide network support at low voltage and distribution level. A single phase, 240 volt, 20kVA prototype was designed and tested and three units installed at Country Energy's research and development facility at Queanbeyan together with three 20kWh battery banks for field trials.

Assessment of the technology continued through 2009/2010 with development and refinement of associated hardware and software to improve the stability and effectiveness of the prototype units.

3. Network Planning

The inverters have ‘four quadrant’ capability, allowing for bi-directional flow of both real (kW) and reactive (kVAr) power. This provides much greater flexibility for the grid interactive inverter in a variety of network based applications aimed at reducing network peak loads, saving network losses (plus associated CO₂), increasing network utilisation and improving quality of supply for customers. Design innovations in switching frequency, control algorithms and assembly techniques promise to deliver substantial efficiency improvements and unit cost reductions relative to standard photovoltaic inverters that have less functionality.

Many of Country Energy’s rural feeders have long sections of Single Wire Earth Return (SWER) or single phase steel conductors where gradual load increases and large point loads impact severely on the quality of supply experienced by customers. Standard network solutions would be to install voltage regulators or rebuild part of the line using a larger conductor, which can run into hundreds of thousands of dollars. An alternative is to use an inverter with a battery bank to store energy during light load periods and release it during peaks to reduce the load on the network. The inverter will also smooth out short-term fluctuations during start up of large loads and reduce voltage rises in SWER systems caused by charging currents during light load periods.

In older, overhead, low voltage networks the original designs were based on much lower household loads and lifestyle changes, such as the increased penetration of air conditioners, can cause similar problems. The normal network response is either to establish a new distribution substation and reconfigure the low voltage or upgrade the low voltage conductors. Grid-interactive inverters providing reactive power to stabilise voltage levels are a viable non-network alternative.

Other applications include:

- **Power factor correction** The grid interactive inverter can provide reactive power to adjust the power factor of the load on the network reducing network loads and losses. The benefits of an inverter over fixed capacitors for this application are that the output is continuously variable and independent of voltage so there are no step changes. The inverter response is instantaneous so the power factor can be constantly adjusted.
- **Short term fluctuations** While the steady state operation of the inverter is limited by thermal ratings the production unit will have significantly higher short term capability to provide ride through support for dips and surges from sudden connection or disconnection of large loads.
- **Improved generation connection capacity** Higher concentrations of embedded generation such as rooftop photovoltaic systems can create problems with voltage rise on the network. The grid interactive inverter can enhance the network’s capacity to absorb generation by providing inductive reactive power to reduce voltage levels.
- **Ancillary benefits** Other benefits observed during field trials include local harmonic suppression and voltage balancing.

Intelligent Network

Country Energy continues to monitor developments in intelligent network technologies and assess the opportunities that these could provide for improved efficiencies in network operation and load management. In particular, intelligent meters and communications can be leveraged to engage customers and influence energy consumption patterns in a way that benefits both the customer and the network. Field trials are proposed during 2010/2011 to evaluate this potential.

Country Energy continues to participate in an international collaboration of utility companies looking at developing intelligent utility networks. Country Energy is actively involved in the Embedded Generation working group of this coalition with the particular aim of facilitating the utilisation of distributed renewable generation and energy storage technologies as an alternative to network augmentation.

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Conductor Replacement Program

Distribution network losses constitute approximately 12 per cent of network demand at peak and, while there is currently no financial incentive for a distributor to invest in losses from an energy perspective, reductions in losses will ultimately have an impact on the need to invest in augmenting the upstream network.

Country Energy has an ongoing program to replace older conductors in order to maintain acceptable levels of reliability and safety. While the core program replaces like for like without increasing network capacity and is not considered to have a demand management outcome, a secondary initiative considers the incremental cost of installing larger conductors, having lower resistance, in order to reduce network demands and energy losses.

While the individual demand reduction contributions are small, the cumulative effect across the network is significant, amounting to an estimated 1,800kW for projects implemented in 2009/2010 and having an associated ongoing CO₂ reduction in the order of 5,000 tonnes per year.

Demand Management Program

Country Energy's Demand Management Program addresses key strategies in the delivery of demand management opportunities within Country Energy in the following key areas:

- Regulatory Compliance
- Program Delivery
- Awareness and Education
- Emerging Technologies
- Stakeholder Communication.

This program includes the assessment of current projects through the demand management process and the development of a Demand Management Program for the regulatory period 2009-2014. This includes both the assessment of projects through the planning process as constraints emerge, which is supported by the extension of the D-Factor Scheme in NSW and the more broadly based demand management programs supported by the Demand Management Innovation Allowance Scheme which commenced in the 2009/2010 financial year.

During 2009/2010 demand management assessments were carried out on seven major projects. All were found to be non-viable due to the likely cost of network support being significantly higher than the avoided cost of the augmentation component of the project.

A consultation paper had been issued during 2008/2009 for the construction of a single 132kV powerline from Mullumbimby to Ballina. An offer to develop a network support option was received and a feasibility study by a non-network service provider was in progress at the end of that year.

The feasibility study subsequently identified partial network support opportunities through existing embedded generators and load curtailment by major customers. The final report determined that the preferred network option should be implemented as the estimated cost of the non-network option, including supplementary temporary generation to provide the shortfall exceeded the avoided cost of the capital project.

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Performance Data

Table 3.5 - Demand management projects implemented during 2009/2010

	Description of demand management project Implemented	Peak demand reduction (kVA)	CO ₂ reduction (tonnes per year and expected duration)	PV of costs of Demand Management Project	PV of total of capital expenditure deferment plus operating expenditure savings
Individual large projects					
1	East Tamworth 7.5MVar capacitor bank	2,250	1,000/20 years	\$640,000	\$1,000,000
3	Macksville 4.5MVar capacitor bank	1,300	250/20 years	\$520,000	\$700,000
4	Boambee South 4.5MVar capacitor bank	1,300	400/20 years	\$520,000	\$700,000
5	Raleigh 4.5MVar capacitor bank	1,000	250/20 years	\$520,000	\$500,000
6	Maclean 7.5MVar capacitor bank	2,000	1,500/20 years	\$640,000	\$1,500,000
Sub-total		7,850	3,400/20 years	\$2,840,000	\$4,400,000
Consolidated projects					
1	Various conductor upgrades throughout Country Energy distribution network	1,800 est.	5,000 est./20 years	Incremental cost estimated at \$10,000/km 304km \$3,040,000	\$3,100,000
2	Load system control upgrades. Estimated 15 per cent of total \$4,900,000 spend) Approx 20,000 relays Finley, Port Macquarie, Lismore, Tweed Heads, Goulburn, Forster	Estimated 1,500 through improved control of off peak load switching	Not significant as only minor reduction in energy usage.	\$735,000	\$3,000,000
3	Capacitor banks	1,200	500/20 years	\$480,000	\$1,200,000
4	Facilitation of PV renewables connection through supply of metering equipment (net community benefits shown)	Est 7,000 (20 per cent of 35,891kW installed capacity coincident with peak summer demand)	47,730 (based on 1.4 MWh per kW installed)/10 years	\$1,770,000 (12,632 sites @ \$140)	\$10,000,000
5	Negotiated outcomes	Negotiations with rural customers to reduce large load connections such as underfloor heating methods on rural feeders			
Sub-total⁷		4,500	5,500/20 years	\$4,225,000	\$7,300,000
Total		12,350	8,900/20 years	\$7,065,000	\$11,700,000

⁷Subtotals do not include the supply of metering equipment to facilitate connection of PV renewable systems.

3. Network Planning

Table 3.6 – Demand management investigations found non-viable in current year

	Description of potential demand management project investigated and reason for non-viability	PV of costs of investigations
1	Ewingsdale 66kV N-1 reliability improvement Not cost effective – RFP resulted in DM feasibility study that identified partial network support potential, through existing embedded generators and load curtailment, to 0.8 MVA winter and 1.3 MVA summer compared to the required 3.7 MVA and 3.9 MVA respectively. Combined cost with temporary installation of supplementary generation (estimated at \$1,350,000) exceeded avoided cost of preferred network option (\$1,000,000)	Estimated \$50,000
2	Finley 66kV N-1 reliability improvement Not cost effective – required 16 MVA network support for growth-related component of \$4,200K	Internal investigation and reasonableness test. Cost estimated at \$5,000 but not separately recorded
3	Uranquinty 66kV N-1 reliability improvement Not cost effective – required 15.5 MVA network support for growth-related component of \$6,420K	Internal investigation and reasonableness test. Cost estimated at \$5,000 but not separately recorded
4	Wilbriggie ZS 33/11kV transformer N-1 reliability improvement Not cost effective – required four MVA network support for growth-related component of \$3,000K	Internal investigation and reasonableness test. Cost estimated at \$5,000 but not separately recorded
5	Bega, Maher Street ZS 66kV bus N-1 reliability improvement Not cost effective – required 16 MVA network support for growth-related component of \$1,500K	Internal investigation and reasonableness test. Cost estimated at \$5,000 but not separately recorded
6	Dubbo, Yarrandale 66kV N-1 reliability improvement Not cost effective – required 25 MVA network support for growth-related component of \$5,220K	Internal investigation and reasonableness test. Cost estimated at \$5,000 but not separately recorded
7	Bathurst, Russell Street ZS, 66/11kV transformer N-1 reliability improvement Not cost effective – required 20 MVA network support for growth-related component of \$626K	Internal investigation and reasonableness test. Cost estimated at \$5,000 but not separately recorded

4. Asset Management

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4. Asset Management

4.1 OVERVIEW

Asset management is a critical component of Country Energy's overall network management strategy and has an important role in determining the outcomes for the business and its customers.

The *NAMP* provides a framework for strategic management of our physical system assets to best support network service delivery. It includes asset management strategies, policies, processes, resources and planned capital investments, asset maintenance and demand management. The *NAMP* is closely related to annual budgets and forecasts for capital, operating and maintenance expenditure planning.

The *NAMP*'s primary objectives include:

- Establishing priorities in line with organisational objectives and statutory obligations, namely safety, reliability and sustainability
- Planning and controlling financing and expenditure in accordance with these priorities
- Ensuring resources are used as effectively and efficiently as possible so that the Government and the community receive the most value for money.

These objectives cover the three major elements of asset management outlined in the *NAMP*:

- Capital investment strategic planning
- Asset renewal and replacement strategic planning
- Asset maintenance strategic planning.

The *NAMP* is designed to comply with the NSW Government's policy on Total Asset Management (TAM). It includes a five-year forecast and reviews capital investment, refurbishment and asset maintenance strategies to ensure a focus on long-term, system-wide and whole-of-life management. These strategies ensure delivery of a secure, high quality, reliable and safe electricity network service that meets the needs and expectations of customers, community, shareholders and other stakeholders at the lowest possible price, and complies with related statutory and regulatory requirements.

Country Energy reviews this plan annually.

4.2 TECHNICAL SERVICE STANDARDS

The Electricity Supply Standards adopted by Country Energy are set out in the document *CEOP8026 Electricity Supply Standard*, in accordance with the *Code of Practice – Electricity Service Standards* published by the former Electricity Association of NSW. The document outlines objectives for technical performance of the network and ensures measures are in place to achieve them. Revised Issue four of this document was released in July 2010.

The main areas addressed include:

- Voltage fluctuations managed in accordance with Australian Standards AS/NZS 61000.3.3:1998, 61000.3.5:1998 and 61000.3.7:2001
- Switching transients (voltage waveform distortion) limited where possible to less than two times normal supply voltage
- Frequency variation and Country Energy's role in notifying AEMO of any sustained fluctuations
- Voltage dips managed through best practice network improvement and augmentation
- Steady state voltage differences between neutral and earth limited to less than 10 volts at the customer's point of supply
- Lightning strikes limited in their impact on supply where possible by adherence to industry best practice system design and maintenance principles

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- Limitation of 'step and touch' voltage differentials managed in accordance with industry standards, namely *ENA Earthing Guide AS/NZS 7000*
- Limiting of voltage imbalance to a six per cent difference on the LV network using 10 min average values between the highest and lowest phase to neutral or phase to phase steady state voltages (this may be exceeded on occasions in rural areas)
- Harmonic content of voltage and current waveforms managed in accordance with Australian Standards AS 61000.3.6:2001
- Mains signalling reliability set at a target of 99.5 per cent failsafe to ensure correct switching and metering functions.

A copy of *CEOP8026 Electricity Supply Standard* can be downloaded from www.countryenergy.com.au

Adoption of Australian Standard AS 60038 – 2000 Standard Voltages

CEOP8026 also outlines Country Energy's adoption of the Australian Standard AS 60038 – 2000 *Standard Voltages*. The standard formally calls for a voltage range of 230V/400V plus ten per cent, minus six per cent, which Country Energy will satisfy by endeavouring to maintain a minimum variance of minus two per cent and an upper bound of plus ten per cent.

4.3 QUALITY OF SUPPLY

Overview

Country Energy monitors its network by various methods including, but not limited to, use of both fixed and portable power quality loggers, electronic tariff meters, maximum demand indicators and portable test equipment. The network is monitored by its system operations department 24 hours a day through the use of various remote system technologies installed in zone substations, field reclosers, meters and various other devices.

Country Energy conducts site specific investigations and monitoring in response to customer-initiated quality-of-supply complaints and enquiries.

Country Energy's specialist power quality technicians, regionally based throughout NSW, conduct site specific investigations and monitoring in response to customer initiated quality of supply complaints and enquiries. They also work with regional network planning staff to proactively monitor network capacity and performance to deliver quality of supply.

Performance data

Country Energy actively participates in the Long Term National Power Quality Survey (LTNPQS), a national power quality survey conducted by the University of Wollongong and a number of other distributors throughout Australia. This survey studies parameters such as steady state voltage, voltage total harmonic distortion (THD), voltage sags and voltage unbalance on three phase sites.

Monitoring by the use of electronic meters at the customers' installation continues to develop. The proposed introduction of electronic meters into in all pad mount substations greater than 200kVA will allow for more in-depth monitoring of network performance.

It is envisaged with the introduction of Smart Meters, an increasing number of customer installations will be able to be monitored in the future enabling the measurement of network performance on a continuous basis. As intelligent networks are developed, a broader view of performance in real time will be available.

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Four power quality disturbance types are analysed in the LTNPQS:

- Steady state voltage
- Voltage unbalance
- Voltage harmonics
- Voltage sags

Due to the quantity of data collated from multiple distributors throughout Australia, the survey results available are taken from the 2008/09 survey period.

The following graphs show trending data results of the four disturbance types analysed in the LTNPQS for Country Energy. For the 2008/09 survey, 55 low voltage and 16 medium voltage sites were analysed.

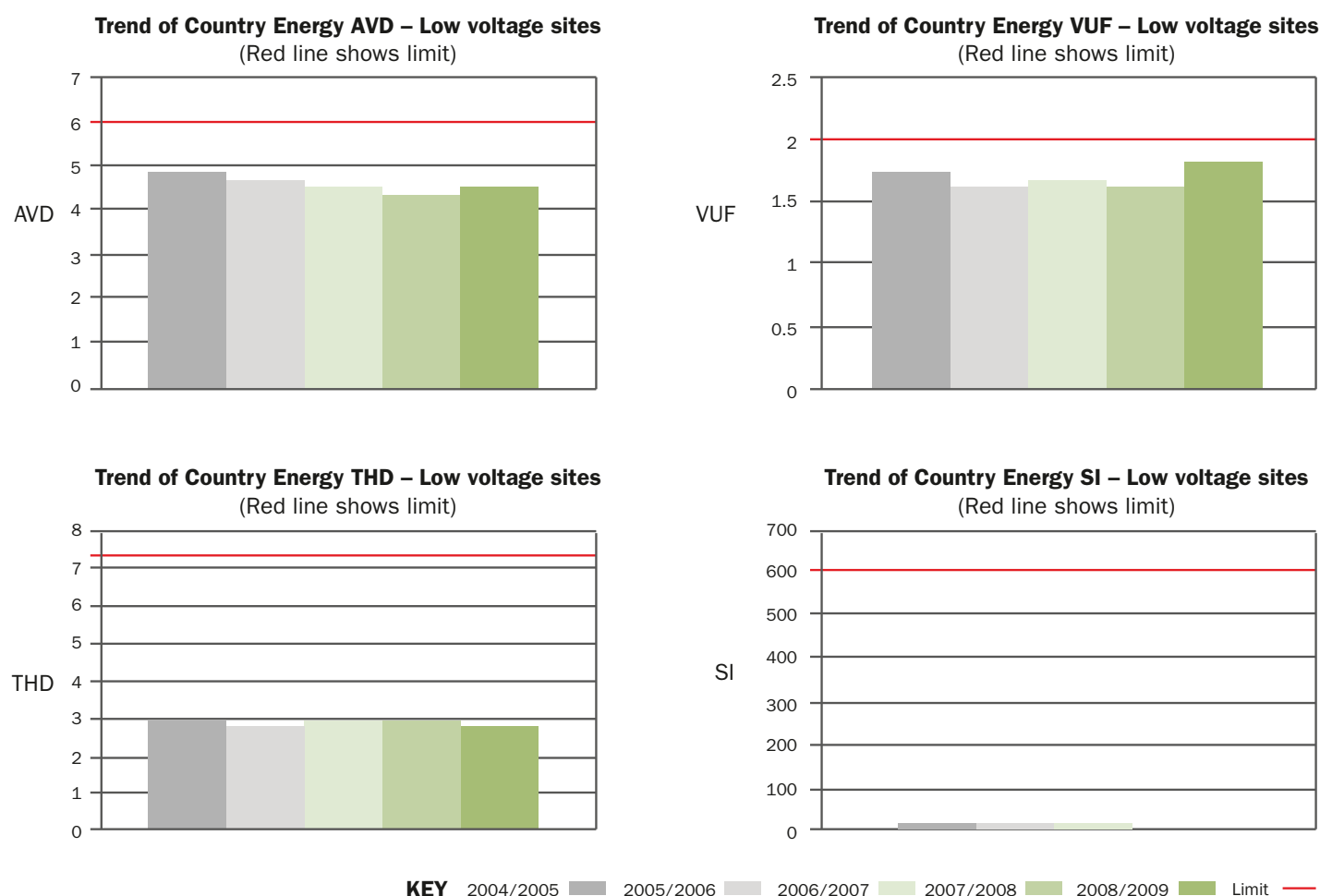


Figure 2 Country Energy's LTNPQS results for low voltage sites

Low voltage median performance indicates unbalance disturbance being the most dominant issue with a value of 92 per cent of the limit, followed by 77 per cent average voltage deviation (AVD). Harmonics and sag data indicates no concerns because the results indicate a trend well below limits.

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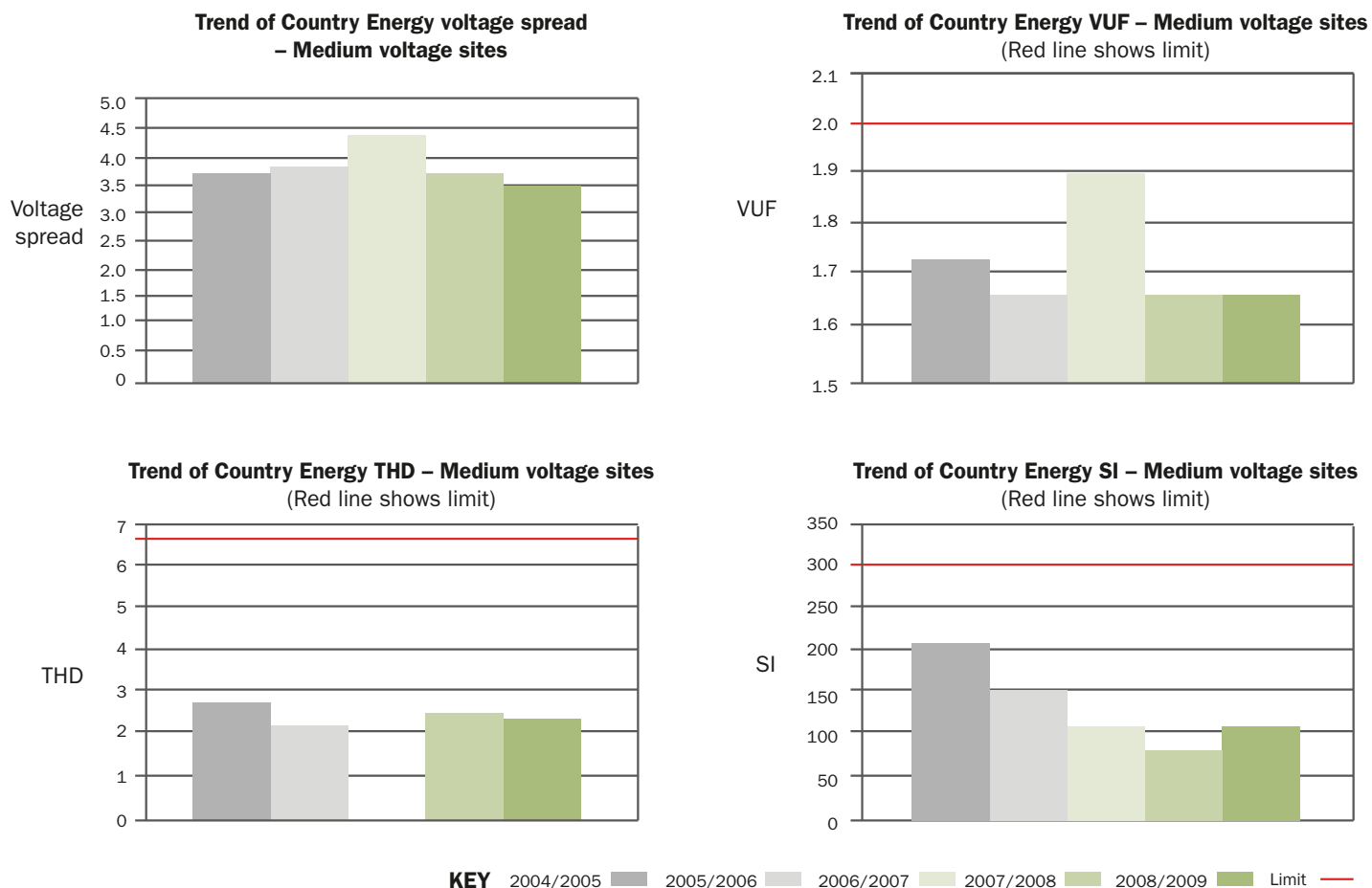


Figure 3 Country Energy's LTNPQS results for Medium Voltage sites

Medium voltage median performance indicates unbalance disturbance being the most dominant issue with a value of 84 per cent of the limit, similar to the low voltage results. Harmonic and sag performance are of no concern as the trend data indicates results well below the limits.

4.4 DISTRIBUTION RELIABILITY

Overview

Country Energy conducts its reliability reporting in accordance with the *NSW Design, Reliability and Performance Licence Conditions*.

Country Energy has 309 urban feeders, 828 short rural feeders and 222 long rural feeders. More than 55 per cent of our customers are on short rural feeders and the average length of these feeders is greater than 60 kilometres.

Implementation of General Electric's ENMAC suite of software was completed on 24 July 2010. ENMAC will achieve an automated interface between systems, producing accurate reliability information to a distribution substation level for outage reporting. Previously, outage data was only recorded to a distribution feeder segment level.

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Organisational detailed performance 2009/2010

Reliability data sets for SAIDI and SAIFI are reported for the whole organisation and feeder categories in Table 4.1.

Table 4.1 – Organisational detailed performance 2009/2010						
Sustained interruption data sets		Whole organisation and feeder category				
Category		ORG	CBD	Urban	Rural Short	Rural Long
Customer Numbers		785,263	N/A	216,842	441,408	127,013
SAIDI	Overall	358	N/A	146	372	673
	Planned	118	N/A	50	117	238
	Unplanned	231	N/A	93	243	427
	Normalised	196	N/A	69	204	384
SAIFI	Overall	2.78	N/A	1.45	3.00	4.31
	Planned	0.50	N/A	0.21	0.49	1.01
	Unplanned	2.11	N/A	1.15	2.29	3.10
	Normalised	1.99	N/A	1.04	2.19	2.88

Organisational performance (normalised) trend

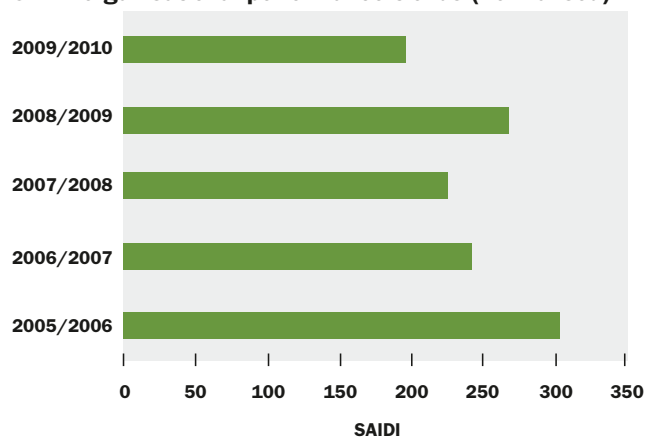
Reliability data for SAIDI and SAIFI (normalised) is reported in Table 4.2 and the graphs show the organisation trends over five years.

Table 4.2 – Organisational performance trends (normalised)					
	Previous years				Current year
	2005/2006	2006/2007	2007/2008	2008/2009	2009/2010
SAIDI	304	242	225	267	196
SAIFI	2.55	2.39	2.28	2.37	1.99

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SAIDI organisational performance trends (normalised)



SAIFI organisational performance trends (normalised)

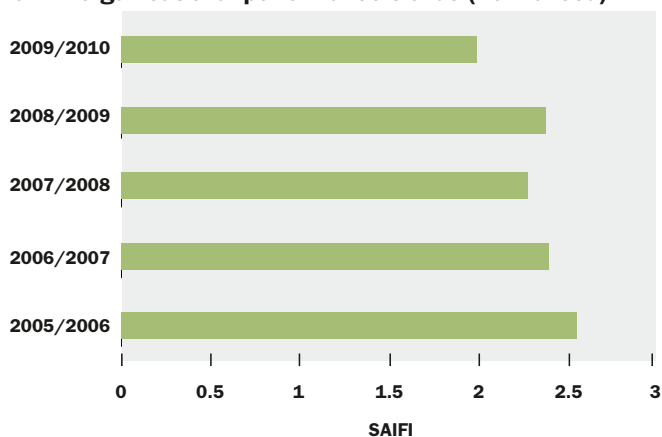


Figure 4 SAIDI and SAIFI performance trends (normalised)

Comment on Performance

Country Energy's network performance during the year is better than mandated in the licence conditions, and a testament to the network improvement strategies that Country Energy is implementing. The lower than normal level of storm activity during the year has also contributed to good performance.

Feeder category performance (normalised) versus targets trend

Table 4.3 – CBD Feeder Performance (Normalised) Trend

		Previous years				Current year
		2005/2006	2006/2007	2007/2008	2008/2009	2009/2010
SAIDI	Actual	N/A	N/A	N/A	N/A	N/A
	Target					
SAIFI	Actual	N/A	N/A	N/A	N/A	N/A
	Target					

Table 4.4 – Urban feeder performance (normalised) trend

		Previous years				Current year
		2005/2006	2006/2007	2007/2008	2008/2009	2009/2010
SAIDI	Actual	109	114	80	110	69
	Target	140	137	134	131	128
SAIFI	Actual	1.28	1.36	1.21	1.36	1.04
	Target	2.00	1.96	1.92	1.88	1.84

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Table 4.5 – Rural short feeder performance (normalised) trend

		Previous years				Current year
		2005/2006	2006/2007	2007/2008	2008/2009	2009/2010
SAIDI	Actual	317	239	233	285	204
	Target	340	332	324	316	308
SAIFI	Actual	2.71	2.47	2.42	2.58	2.19
	Target	3.30	3.24	3.18	3.12	3.06

Table 4.6 – Rural long feeder performance (normalised) trend

		Previous years				Current year
		2005/2006	2006/2007	2007/2008	2008/2009	2009/2010
SAIDI	Actual	578	497	431	483	384
	Target	750	740	730	720	710
SAIFI	Actual	4.06	3.82	3.50	3.47	2.88
	Target	5.00	4.90	4.80	4.70	4.60

Comment on performance

Country Energy's network performance during the year is better than mandated in the *licence conditions*, and a testament to the network improvement strategies that Country Energy is implementing. The lower than normal level of storm activity during the year has also contributed to this good performance.

Excluded Events

Table 4.7 – Excluded Interruptions for 2009/2010

Date of Event	Description of Event	Number of Customers Interrupted	Maximum Duration of Interruption (minutes)	Effect of Event on SAIDI Figure (minutes)	Basis for Exclusion
2 September 2009	132kV Transformer Failure	36,437	542	8.41	Exceeded Tmed threshold
28 January 2010	Storm Activity	44,947	1,562	9.82	Exceeded Tmed threshold
30 May 2010	Storm Activity	16,968	3,676	18.53	Exceeded Tmed threshold

Major event day TMED

The value of TMED for 2009/2010 was 6.44 minutes.

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Individual feeder reliability performance

The performance objectives for organisational average performances for each feeder category are not sufficient to identify when customers on a particular feeder experience unsatisfactory reliability performance. For this reason, SAIDI and SAIFI criteria (after 'excluded interruptions' are disregarded) act as a trigger for investigation and exception reporting purposes. The figures contained in the ministerially imposed *licence conditions* are shown in Table 4.8.

Table 4.8 – Individual feeder standards for exception reporting 2009/2010

Category	Feeder categories			
	CBD	Urban	Rural short	Rural long
SAIDI	N/A	400	1,000	1,400
SAIFI	N/A	6	8	10

Performance outside this range results in the following actions:

- Immediate investigation of the causes for each feeder exceeding the individual feeder standards
- By the end of the quarter following the quarter in which the feeder first exceeded the standard, completion of an investigation report identifying the causes and action required to improve the performance
- Completion of any operational actions identified in the investigation report by the end of the third quarter following the quarter in which the feeder first exceeded the standard
- Where the investigation report identifies actions, other than operational actions, there is a requirement to improve the performance of each feeder to the individual feeder standards, develop a project plan, including implementation timetable and commence its implementation by the end of the second quarter following the quarter in which the feeder first exceeded the individual feeder standards.

Summarised performance against the above *licence conditions* is shown in Table 4.9.

Table 4.9 – Individual feeder performance against the standard summary

Category	Feeder type			
	CBD	Urban	Rural short	Rural long
Feeders (total number each type)	N/A	309	828	222
Feeders that exceeded the standard during the year (total number)	N/A	14	76	33
Feeders not immediately investigated (total number)	N/A	-	-	-
Feeders not subject to a completed investigation report by due date (total number)	N/A	9	41	19
Feeders not having identified operational actions completed by due date (total number)	N/A	2	8	4
Feeders not having a project plan completed by due date (total number)	N/A	2	14	9

Comment on performance

Following periods of high network activity, an increased number of feeders required investigation. In all cases, the investigation commenced on time and interim remedial actions were taken where appropriate, even though the report may not have been completed in the required timeframe.

The number of feeders exceeding individual feeder standards is reducing and currently is nine per cent of the total number of feeders. Country Energy has implemented processes to expedite investigations.

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4.5 TRANSMISSION RELIABILITY

Transmission reliability performance data

Country Energy has no requirement to provide this data.

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5.1 OVERVIEW

Country Energy's goal continues to be Australia's Safest and Healthiest essential service provider while being a vital part of the communities, with a strong emphasis on being an environmental leader.

The aim is to integrate safety, security, health and environment into all that Country Energy does.

To demonstrate commitment to safety, security, health and environment Country Energy will:

- Strive to be an incident-free organisation
- Provide a safe, secure, healthy and environmentally conscious working environment through the effective implementation of this policy
- Apply a risk management approach to the full scope of activities, products and services to reduce hazards
- Comply with relevant legislation, regulations, standards, codes, licences and commitments
- Ensure all employees are trained and have the knowledge and skills they need to undertake their work in a safe, secure, healthy and environmentally responsible manner
- Require supervisors, employees, contractors, labour hire employees and visitors to abide by all SSHE policies, procedures and other requirements
- Engage in effective consultation and open communication about SSHE issues with employees, employee representatives, contractors and labour hire employees
- Conduct incident investigations fairly, with a focus on implementing preventative actions
- Ensure appropriate resources are applied to fulfil the aims of this policy
- Establish measurable objectives and targets based on our significant SSHE hazards to continually review and improve safety, security, health and environmental performance
- Minimise the impact of pollution generated by activities on the environment by reducing emissions, discharges and wastes by promoting energy conservation and recycling of wastes
- Promote a 'Safety First' culture where everyone watches out for their workmates, families and local communities.

Performance highlights

Major progress has been made in the areas of Safety Security Health and Environment during the past year within Country Energy. Achievements over the past year have been:

- Continued certification to Australian Standard AS/NZS 4801:2001 *Occupational Health and Safety Management Systems*
- 1,131 safety process audits conducted, focusing on premises, work teams and individuals achieving an average score of 98.3 per cent
- Lost time injury frequency rate of 3.5
- Increased focus on contractor management
- The organisational Safety Policy has recently been reviewed and rewritten, resulting in the all-encompassing Safety, Security, Health and Environmental Policy
- Continued effort to find new and better ways in addressing the most critical areas of our Safety, Security, Health and Environmental (SSHE) management system, through consolidation and review of Country Energy's internal processes, resulting in a new easy-to-use manual
- Comprehensive regional safety plans in action across the business; a strong commitment to attendance and contribution at training: corporate induction, practising lineworker, regulatory assessment days and Skilling Up sessions
- Increased awareness of manual handling issues throughout the business through the Manual Handling Solutions Program

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- A continued emphasis of the Powerful Minds Program to conduct risk assessments of psychological issues in the workplace. In-house training has included 223 employees who have attended the 12-hour Mental Health First Aid course, 69 managers and supervisors who have attended one day Powerful Minds training
- The Powerful Apprentice Pilot has identified a number of wider recruitment, training, and mentoring opportunities that are now being addressed by a comprehensive Better Apprentice Program for the future
- Development of comprehensive public safety fact sheets that are now available on the website
- Recognised positive achievements in safety performance and culture, with 93 sites successfully achieving the milestone of five years or greater without a lost time injury, compared with 85 sites in the previous year
- Country Energy was a finalist in three categories of the NSW Safe Work Awards 2009 – Best Workplace Occupational Health and Safety Management System; Best solution to an identified workplace safety issue for the manual handling solutions program; and Alan Burman, Manager Public Safety, was the winner for best individual contribution to workplace health and safety.

Safety for our people – Programs

Occupational health and safety and Environmental Management System

Country Energy's Occupational Health and Safety and Environmental Management System is encompassed in a corporate guideline known as the Safety, Security, Health and Environmental (SSHE) Management System Manual. This manual provides the framework for all systems and activities within the organisation, reduces duplication and sets a platform for consistency of documentation and has a focus on continuous improvement and quality management.

During 2009/2010, two independent surveillance audits of the OHSEMS were again conducted by NCS International Pty Ltd. The outcome of these resulted in continuing certification to *AS/NZS 4801:2001 Occupational Health and Safety Management Systems* and *ISO 14001:2004 guideline for an Environmental Management System*.

Workers Compensation 2009/2010

The focus this year has been on trying to maintain early reporting statistics in light of a reduced time frame for all injury reporting by WorkCover to 48 hours. This requirement was met 75 per cent of the time and maintained a 98 per cent reporting within five days, which was the previous reporting requirement.

We have also undertaken a review of processes in order to qualify for a WorkCover sponsored change of calculating premium (Retro Paid method). This change takes effect on 1 July 2010.

Overall, costs are within our target range and the return to pre-injury duties rates remains strong. Country Energy also recently qualified for a grant (\$121,000) from the scheme insurer Employers Mutual Limited, for a behavioural safety program to be undertaken over the next 12 months – (see below).

Creating a safe work culture

Country Energy has undertaken a range of activities that have contributed to substantial improvements in safety performance over the past five years. However, the commitment to zero harm for employees requires Country Energy to continuously look for opportunities to improve. Monitoring of incident trends and causal factors continues to highlight human factors as a key area of concern.

Country Energy has successfully obtained a \$121,000 grant in partnership with insurer Employers Mutual Limited to build upon programs addressing human risk management in the workplace. These programs offer another avenue to pro-actively prevent incidents and improve safety culture and behaviour. The program will reference best practice initiatives from like-minded industries, international research and consultation with the workforce. This holistic approach to safety compliments a mature safety management system and recognises Country Energy's stance as a highly reliable organisation.

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TotalSAFE

The Safety Incident and Hazard Reporting support system (TotalSAFE) has been used across Country Energy now for two years. The first year saw all audit and inspection results being recorded and to date there is an excess of 39,000 records. By having audits recorded electronically Country Energy has been able to analyse the results; this provides an opportunity to develop and improve the audit process. In the areas of the Safety *and* Environmental Audit and Safety *and* Environmental Field Audit Country Energy is introducing new audit templates; new questions and a renewed focus on continuous improvement.

Country Energy's entire incident reporting has been successfully moved across to TotalSafe by the second year of operation. This has provided greater visibility with reports available immediately, as well as enabling easy access to the management of incidents.

Ongoing and future enhancements

Powerful Health

Powerful Health is Country Energy's workplace health promotion program. During 2009/2010, the program offered a range of activities including influenza vaccinations, skin cancer subsidy, \$99 subsidy towards health and fitness activities, \$500 grant subsidy for team activities, 10,000 steps program, corporate health events and a range of regional health initiatives such as Health Field Days.

During the year, the program took a major focus on promoting team based activities to increase physical activity levels but also promoted team building and healthy competition. The range of group activities extended from boot camp sessions to meditation workshops, charity cricket matches and corporate team fun run events.

A major Powerful Health campaign run in 2009/2010 was the launch of influenza vaccinations on site as part of World Day for Safety and Health at Work. Additional activities run in conjunction with flu vaccinations included healthy morning tea, health and wellbeing information stands, and seated massage.

Powerful Minds

Country Energy is committed to both the physical and psychological safety of its employees, and for this reason continues to develop a risk management approach to psychological health through the Powerful Minds program. An ongoing focus of the program during 2009/2010 has been to increase mental health literacy and promote an early intervention approach to identified symptoms of stress in the workplace. Eleven Mental Health First Aid courses were run by in-house trainers, with 223 employees participating. Follow-up Powerful Minds sessions were also conducted with 69 managers and supervisors.

A major focus of this year was the delivery of Powerful Teams sessions for all employees in the North Western region of Country Energy. The session provided information on mental health issues, substance abuse, and bullying and harassment. Evaluation data from the NW sessions provided in Figure 5 and Figure 6.

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Increased understanding of discrimination, harassment and bullying

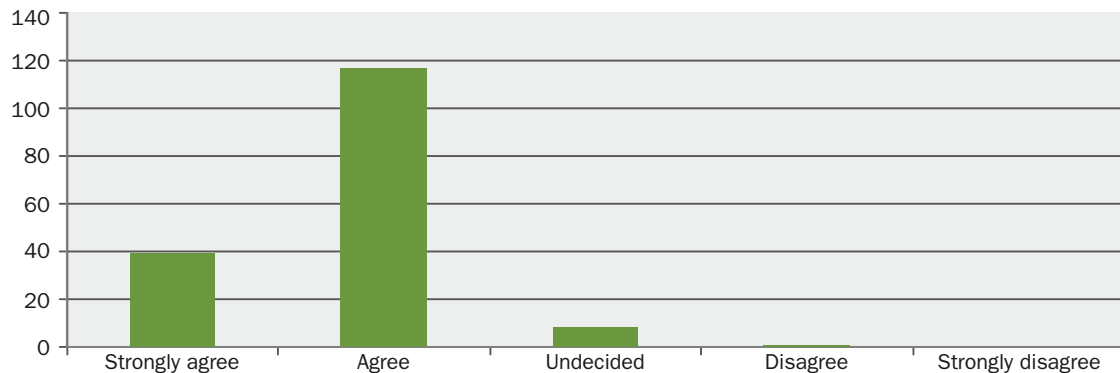


Figure 5 Understanding of discrimination, harassment and bullying

How effective did you find this method of training in workplace discrimination, harassment and bullying?



Figure 6 Effectiveness training to prevent discrimination, harassment and bullying training

Manual Handling Solutions Program

Manual handling is a workplace hazard that causes injury. Within Country Energy, the majority of workers compensation claims are for musculoskeletal injuries that can be attributed to manual handling. The Manual Handling Solutions Program provides a risk management framework for employees to develop solutions to these hazards with a strong emphasis on consultation and communication.

A comprehensive evaluation of the program since its launch in 2007 has demonstrated a number of solid outcomes:

- A reduction in incidents and injuries resulting from manual tasks has been observed since the launch of the Manual Handling Solutions Program in 2007. The most significant impact on *incident rates* and *cost of claims* occurred in the early stages after the launch
- Manual handling *claims numbers* and *rate of claims* have continued to steadily decline
- *Injury severity* and resulting workers compensation *claims costs* may have been positively impacted by expansion of an in-house team of health professionals providing injury management expertise

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- Manual handling assessment is embedded in the Skilling Up leadership program for supervisors and has resulted in 36 improvement suggestions. The 2009 Senior Development Program participants have also engaged in two manual handling projects. These proactive safety leadership initiatives are important to the development of a sustainable solutions-focused culture.

Introduction of manual handling criteria to the field audit tool is also expected to drive implementation of solutions and, combined with coaching, offer a platform for continuous improvement. The program provides a solutions-focused approach to a recognised hazard. For the duration of the program:

- 210 manual handling improvements have been suggested by Country Energy employees since the program launch; 36 of these were made in response to the Skilling Up program
- 120 manual handling assessments have been undertaken; 29 of these progressed to manual handling solutions, with 63 works in progress
- The most significant hazards are generally targeted for assessment. A focus on these hazards is important to retain as priority areas of concern
- The program facilitates partnerships with different areas of the business to achieve common goals. Joint initiatives with Property, Fleet and Design and Construction Standards are contributing to a number of improvements across the business.

The manual handling program has sought to address office based risk factors related to work practice at the computer workstation, through the trial of the Workspace/Wellnomics rest break software program. Further roll out of the program is now planned for the business.

The Manual Handling Solutions Program has received industry recognition, through representation at external conferences, Safe Work awards and industry field days. These accomplishments recognise the substantial and wide ranging contribution to the program by Country Energy employees.

Powerful Apprentice Program

The Powerful Apprentice Program is a four-year pilot program tracking health and safety issues faced by current third year apprentices within Country Energy. Third year apprentices who attend the Goulburn Training Centre have been in the spotlight for the past three years, so the Safety and Employee Services Team can better understand the issues apprentices face, and develop programs to support all future apprentices.

A critical focus of the program in 2009/2010 has been to address functional stability issues and psychometric testing. With the Learning and Development Team and Technical Training Services becoming part of Safety and Employee Services during this year, an even more comprehensive and co-ordinated approach has been taken towards apprentice safety. The Powerful Apprentice Pilot identified a number of wider recruitment, training and mentoring issues that are now being addressed by a Better Apprentice Program for the future. As well as the practical training elements apprentices, it is important that the Better Apprentice Program addresses social and cultural issues for our apprentices by better understanding them as a group; the better they can be supported them during this critical training time of their lives.

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HealthWatch

Health Watch is the Country Energy health surveillance program. The primary aims of the program are to:

- Achieve regulatory and legislative compliance in regards to health surveillance requirements
- Identify key hazardous exposure activities and implement appropriate controls, that will ensure long-term health and safety improvements for Country Energy employees
- Enhance management of the data collection, storage and retrieval process for health surveillance information.

A primary focus of the program in 2009/2010 to achieve these aims has been the revised development of the TotalSAFE OHS management pages to meet the recording and reporting needs of health surveillance. Data from medical questionnaires completed by 54 per cent of our employees will be analysed for trending issues across the business, to better ensure the long-term health of employees from occupational exposures.

5.2 SERIOUS ELECTRICITY NETWORK ACCIDENTS (PUBLIC)

Country Energy's public safety program has achieved extraordinary results over its lifetime. The combination of targeted advertising and the direct intervention approach has been very effective. As a result, the number of reportable incidents remains low with only 11 occurring this year. Although this is a slight rise of one from the 2008/2009 year it is in the light of a significant increase in agricultural activity after some good rains.

This success rate makes Country Energy a world leader in the field of public safety and, with the implementation of a number of world firsts – including the overhead line flag marker – and continuing development of new approaches, Country Energy will remain at the forefront of public safety.

Country Energy's public safety manager recently won an award from WorkCover NSW for 'Outstanding contribution' and was a finalist in the national Worksafe Australia awards partially based on his contribution to Public Safety within Country Energy.

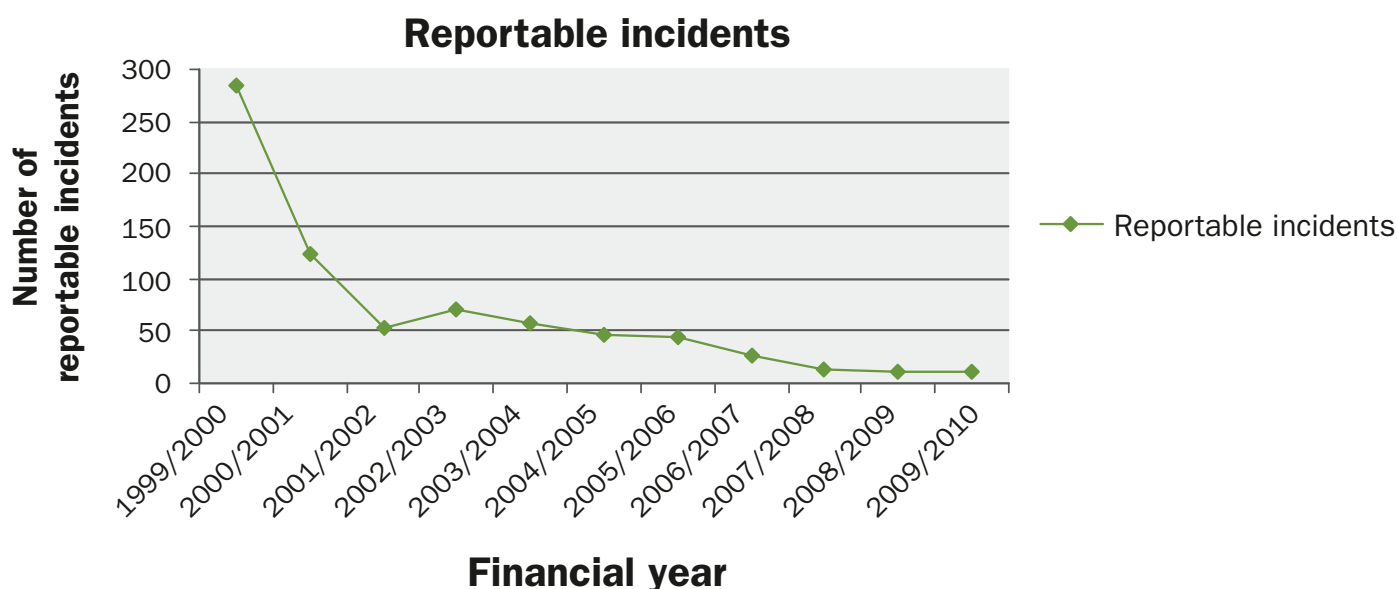


Figure 7 Reportable serious electricity network accidents (public)

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While Country Energy is proud of this achievement, it will continue to strive towards achieving a target of zero.

Public safety strategy is aimed at identifying 'at risk' groups and working with them to provide targeted educational and practical assistance in managing electrical safety issues – and to provide public safety training free of charge to the main 'at risk' groups.

During the year, Country Energy worked with emergency services, earthmovers, construction workers, cotton growers, pilots, transport groups and local and state government workers. Surveys of these groups have shown 90 per cent retention of the main educational messages and information – an excellent achievement.

Country Energy has increased our public safety resources from four to nine to assist in continuing to target key industries such as agriculture, aircraft and construction operations.

Electrical Hazard awareness sessions have been delivered to many different groups across and outside its footprint including:

- Cotton harvesting companies
- Local councils
- NSW Rural Fire Service
- Earth moving companies
- Schools
- Pilot clubs.

Electrical hazard awareness training has also been delivered to organisations outside Country Energy's footprint, as much of the activity occurs in its area.

Table 5.1 – Serious electricity network accidents (public) trend

Category	Previous years				Current year
	2005/2006	2006/2007	2007/2008	2008/2009	2009/2010
Non-Fatal	1	2	2	1 ⁸	2
Fatal	2	2	–	–	2

⁸Only includes incidents that are required to be reported under ENPR guidelines.

Fatal Incidents

- A 77-year-old ACT man contacted powerlines with his light aircraft apparently while landing on an improvised airstrip. The presence of the strip was not known to CE and the powerlines were directly in the approach path. The incident occurred on 2 December 2009 near Frogmore.
- A farmer sustained apparent fatal electric shock after making contact with a 11kV powerline. This contact was made with an aluminium water pipe measuring 9.4 metres in length which the farmer was attempting to manoeuvre into position.

Non- fatal incidents

- A member of the public came into contact with 22kV. The person drove into a paddock with a boom spray. He then got out of his vehicle and proceeded to unfold the wings of the boom spray unit. It was at this point that one of the wings came into contact with the 22kV overhead conductors.
- A member of the public gained unauthorised access to a distribution substation and received an electric shock

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5.3 ACTIONABLE ELECTRICITY NETWORK SAFETY INCIDENTS (PUBLIC)

Public safety incidents continue to decline as a result of the strategies in place to address most types of incidents. In the coming year Country Energy will be focusing on aircraft, with a series of new strategies currently being developed. Aircraft incidents form a disproportionately high number of the incidents that occur and a number of internal workshops are being held to broaden the approach to solutions.

Country Energy is carrying out research and development to pro-actively manage public safety risk arising from poor neutral conductor connections. Every Country Energy employee has been issued one of these devices as part of this trial.

Country Energy is also working with a meter manufacturer to have neutral integrity functionality included in Smart Meters. Once the technology is proven, any roll out of these Smart Meters and an associated communications network in the future will allow Country Energy to remotely monitor this important public safety issue.

Table 5.2 – Actionable safety incidents (public) on the electricity network classified by network element trend

Category of incident	Previous Years				Current Year
	2005/2006	2006/2007	2007/2008	2008/2009	2009/2010
Overhead mains in position	35	11	9	9	7
Overhead mains fallen	4	4	4	-	-
Overhead service in position	7	-	-	-	-
Overhead service fallen	-	2	-	-	-
Underground mains	1	1	-	-	2
Underground service	-	-	-	-	-
Street light fitting or support	-	-	-	1	-
Cable boxes, pillars and service cabinets	-	-	-	-	-
Substation (excluding pole-type) – Fire and/or explosion	-	-	-	-	-
Substation (excluding pole-type) – Other	-	-	-	-	1
Substation (pole-type) – Fire and/or explosion	-	-	-	-	-
Substation (pole-type) – Other	-	-	-	-	-
Poles columns or towers	-	1	-	-	1
Air break switches	-	-	-	-	-
Pole mounted equipment (reclosers, drop out fuses, etc.)	-	-	-	-	-
Service connection or customer switchboard	-	1	-	-	-
Other (describe below)	1	-	-	-	-
Total	48	26	13	10⁹	11

⁹Only includes incidents that are required to be reported under ENPR guidelines

5. Network Safety

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5.4 SERIOUS ELECTRICAL NETWORK ACCIDENTS – NETWORK WORKER AND ACCREDITED SERVICE PROVIDER

A summary of performance, trends, causes and preventative action implemented are shown in Table 5.3. The network operator contestable work ring fenced operation is shown in the accredited service provider section of Table 5.3.

Table 5.3 –Serious Accidents (Network Worker and Accredited Service Provider) on the Electricity Network Trend

	Previous Years								Current Year	
	2005/2006		2006/2007		2007/2008		2008/2009		2009/2010	
	NF	F	NF	F	NF	F	NF	F	NF	F
Network Operator Employees	6	1	1	-	4	0	1 ¹⁰	-	-	-
Network Operator Contractors	-	-	-	-	-	-	-	-	-	-
Accredited Service Providers	-	-	-	-	-	-	-	-	-	-

¹⁰By definition, this should not be reported as a serious accident and it should only be reported as an actionable safety incident.

5.5 ACTIONABLE ELECTRICITY NETWORK SAFETY INCIDENTS (NETWORK WORKERS)

The continued decline in incidents from 2007/2008 can be attributed to the success of the Practising Powerline Workers course along with an increased focus on safety process auditing, mentoring and coaching.

The Practising Powerline Workers course merged the technical knowledge and skills of the new apprentice powerline workers with the ageing qualified line workers, using this connection to make fundamental changes to behaviours and challenging acceptance of poor work practices.

The overall objective to enhance professional, technical and behavioural competencies of Country Energy's powerline workers through quality training/mentoring and education has been successful.

5. Network Safety

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Table 5.4 – Actionable safety incidents (network workers) on the electricity network classified by network element trend

Category of Incident	Previous years				Current year
	2005/2006	2006/2007	2007/2008	2008/2009	2009/2010
Overhead mains in position	7	15	5	2	-
Overhead mains fallen	1	2	-	-	-
Overhead service in position	2	-	-	1	-
Overhead service fallen	-	-	-	-	-
Underground mains	2	-	3	-	-
Underground service	-	-	-	-	-
Street light fitting or support	1	1	-	-	-
Cable boxes, pillars and service cabinets	1	-	-	1	-
Substation (excluding pole-type) – Fire and/or explosion	-	-	-	-	-
Substation (excluding pole-type) – Other	-	-	1	-	-
Substation (pole-type) – Fire and/or explosion	1	-	-	-	-
Substation (pole-type) – Other	-	1	-	1	-
Poles columns or towers	-	-	-	-	-
Air break switches	-	2	-	-	-
Pole mounted equipment (reclosers, drop out fuses, etc.)	1	1	-	-	-
Service connection or customer switchboard	-	2	-	-	-
Other (describe below)	-	-	-	-	-
Total	16	24	9	5	-

5. Network Safety

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5.6 MAJOR INCIDENT REPORTS

Network operators are required under the *licence conditions* to report to the Minister for Energy, within 24 hours, any major network incidents involving significant injury to persons, loss of property or widespread supply interruptions. High level severity incidents are to be advised immediately.

Network operators shall list all major incident reports made to the Minister as required under these *licence conditions*.

Table 5.5 – Summary of major incident reports submitted for 2009/2010

Date	Incident Description	Locality
07/07/2009	Supply lost to Coonamble, Gilgandra, Gulargambone	Gilgandra
17/07/2009	Supply lost to Egansford zone substation - car hit pole	Colleambally
03/08/2009	Supply lost to Orange west zone substation	Orange
09/08/2009	South Tamworth zone substation tripped	Tamworth
29/02/2010	South Tamworth zone substation tripped	Tamworth
12/08/2009	Forbes – crane contact	Forbes
11/08/2009	Cowra – 17 hours off for 11 customers	Cowra
26/08/2009	Failure of 66kv HV Tie, Crookwell, Goulburn	Goulburn
25/08/2009	Tripping of 66KV line due to debris	Wagga Wagga
03/09/2009	Tripping of Dubbo 132 KV (suspect CB)	Dubbo
21/09/2009	Supply lost to Broken Hill	Broken Hill
19/09/2009	Supply lost to Wellington 132KV and Mumbil	Wellington
21/09/2009	Supply lost to Broken Hill (three x SENI- 2'S)	Broken Hill
30/09/2009	Supply lost to Bodalla, Moruya Town, Tuross substations	Moruya
29/09/2009	Brogo zone substation	Brogo
29/10/2009	Supply lost to Young zone substation	Young
31/10/2009	Balloon in 22kv powerline	Murrumbateman
10/11/2009	Transformer tripped at Narromine	Dubbo
04/04/2010	Coffs Harbour 132KV	NSW North Coast
12/05/2010	North Coffs – 66KV Fault	NSW North Coast
03/06/2010	Lennox Head (Cyclone)	NSW North Coast
29/02/2010	South Tamworth zone substation tripped	Tamworth
20/03/2010	Boronia St zone substation	Port Macquarie
16/05/2010	Tractor hit Pole	Mid North Coast
30/05/2010	10 X Seni-2 forms for storms on NSW South Coast	South Coast
12/05/2010	66KV Conductor down at Tumut 132KV	Tumut
25/05/2010	66KV aerial earth failed – Wagga Wagga	Wagga Wagga
02/12/2009	Fatal incident where a pilot in a light aircraft contacted powerlines	Frogmore
29/09/2009	Fatal incident where a farmer moving water pipe contacted an 11kV powerline	Tenterfield
28/08/2009	Member of the public injured when a boom spray unit contacted 22kV powerline	Tallimba
08/04/2010	Member of the public injured after gaining unauthorised access into a distribution substation	Albury

6. Customer installations

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6. Customer Installations

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6.1 OVERVIEW

Monitoring of installations is carried out in accordance with regulations and industry codes of practice. In addition to the *Electricity Supply (Safety and Network Management) Regulation 2008*, installations must be consistent with the *Electrical Safety (Electrical Installations) Regulation 1998*, the *Code of Practice for Installation Safety Management* and the *Code of Practice for Contestable Works Accreditation*.

Country Energy's *CEPG8004 Customer Installation Safety Plan* outlines the approach taken to managing all aspects of customer installation work. The plan defines our approach with respect to the roles of both contractors and Country Energy.

Installation and inspection data is managed within Country Energy's Contractor Performance Register which enables management of CCEW and NOSW information and the auditing of individual contractors to ensure adherence to *CEPG8004*.

6.2 REPORTS AGAINST CUSTOMER INSTALLATION SAFETY PLANS

Table 6.1 - Installation Inspections Trend

	Previous Years				Current Year
	2005/2006	2006/2007	2007/2008	2008/2009	2009/2010
Number of Notifications (CCEW)	23,387	20,651	19,721	18,771	26,614
Number of Inspections	11,221	9,404	8,555	7,297	10,920
Installation Inspection Rate (%)	48.0	45.5	43.4	38.9	41
Major Safety Defect Rate (%)	3.0	3.3	2.4	2.8	2.5
Safety Breach Notices Issued (%)	11.8	3.4	2.5	2.9	2.6
Number of Warnings Issued	306	323	212	214	280
Reports to NSW Fair Trading (No.)	7	5	5	4	5
Number of Audits by Network Operator	354	68	37	28	91

6.3 CUSTOMER INSTALLATION SHOCK REPORTS

It is pleasing to report that the downward trend of reported public shocks has continued in 2009/2010 with a further 5.7 per cent reduction over the previous period, a total of around 19 per cent for the past two years.

Also delivering positive outcomes is the project involving the replacement of targeted overhead services. This initiative has reversed the upward trend in the 'Faulty Service Joint' category with a 35 per cent reduction in shocks compared to the previous year's figures.

The number of reported incidents in the 'Lightning/Storm' category has decreased considerably over the past 12 months. This trend is directly related to the lower number of severe weather events that occurred during the summer storm season.

'Faulty mains box' incidents continue to rise with a 16 per cent increase reported in this period. It is likely that most customers would not be aware that the mains box is part of the customers' installation and that they have the responsibility for maintaining this equipment. The subsequent lack of maintenance could be directly related to the upward trend in this category.

6. Customer Installations

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Table 6.2 - Customer installation shock reports trend

	Previous years				Current year
	2005/2006	2006/2007	2007/2008	2008/2009	2009/2010
Shocks on Customer's Premises (Number Reported)	696	608	834	719	678

Table 6.3 Customer installation safety – Categories of shocks analysed for 2009/2010

Category	Number		% of total
	Fatal	Non-fatal	
Contact with consumers' mains – faulty mains	–	6	0.9
Contact with consumers' mains – human error	–	4	0.6
Contact with live parts at switchboard – faulty switchboard	–	1	0.1
Contact with live parts at switchboard – human error	–	11	1.6
Unsafe installation work by licensed contractor	–	17	2.5
Failure of part of installation (not water related)	–	13	1.9
Defective or unsuitable appliance	–	27	4.0
Working on or interference with installation	–	12	1.8
Working on or interference with appliance	–	2	0.3
Water damage or ingress	–	7	1.0
No cause found (including static electricity)	–	64	9.4
Other (described in table 6.3A)	–	514	75.8
Total	–	678	
Total per 1,000 customers	–	0.845	

6. Customer Installations

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Table 6.3a - Categories of 'other' shocks analysed for 2009/2010

Category	Number		% of total
	Fatal	Non-fatal	
Cause category installation related			
Faulty mains box	–	65	12.6
Poor earthing	–	15	2.9
Other	–	78	15.2
Subtotal	–	158	
Cause category network related			
Faulty mains joint	–	65	12.6
Faulty service joint	–	85	16.6
Faulty underground joint	–	19	3.7
Long lv run	–	24	4.7
Hv leakage	–	2	0.4
Incorrect polarity	–	2	0.4
Other	–	45	8.8
Sub total	–	242	
Other cause categories			
Lightning/storm	–	11	2.1
Nuisance tingles <10 volts	–	46	8.9
N/a	–	1	0.2
Subtotal	–	58	
Undefined (under investigation)	–	56	10.9
Total			
	–	514	

7. Contestable works Scheme

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7. Contestable works Scheme

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7.1 OVERVIEW

Country Energy continues to use the Contractor Performance Register to monitor all level two works. Level one and level three work is collected regionally by the quality control coordinators, working in conjunction with the regional planning and customer connection managers.

2009/2010 has seen the following trends:

- **Level one:** Internal project notifications are down and inspection rates falling in line with this decline. This is partly explained by the increase in external Level one works notices and inspections.
- **Level two:** The 2009/2010 trends in project notices, particularly in the last six months, has been attributed to the industry's involvement in the solar bonus scheme. The rigour of Country Energy's commitment to the metering, commissioning and inspection of these installations has created the need for additional resources across all the respective business units. The additional resources have enabled Country Energy to retain the previous year's inspection average and confirm the increased involvement by all the stakeholders as being constructive with the industries desired standard being achieved.
- **Level three:** A significant increase in certified external designs is reflective of the increase in Level one external works. As more external Accredited Service Providers continue to gain accreditation to operate within the service area this increase will continue to meet demand of contestable market.

Table 7.1 – Contestable works trend

	Previous years								Current year	
	2005/2006		2006/2007		2007/2008		2008/2009		2009/2010	
	Int	Ext	Int	Ext	Int	Ext	Int	Ext	Int	Ext
Network work (Level 1)										
Project notifications	2,336	396	1,861	799	1,520	1,103	778	1,269	538	1,434
Initial inspections of completed projects	1,729	414	1,048	618	540	623	786	947	595	907
Of projects inspected, number initially nonconforming	389	45	175	85	109	146	220	200	46	159
Customer connection work (Level 2)										
Notifications (NSW)	6,780	13,028	6,299	13,203	5,573	13,238	5,518	17,135	3,338	19,910
Inspections by network operator	4,321	9,456	3,540	9,268	3,143	11,807	3,485	12,919	2,526	12,433
Major defects	6	117	3	130	-	102	3	107	2	62
Network design work (Level 3)										
Designs certified	2,336	410	1,861	631	1,225	1,114	778	947	538	1,234

8. Bushfire risk management

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8. Bushfire Risk Management

8.1 OVERVIEW

Country Energy is committed to achieving effective asset management by implementing strategies to ensure the safe and reliable operation of the network. A key component is to manage assets in a manner that minimises the risk of bushfires, protects assets and maintains the highest possible supply reliability during bushfire risk periods.

While Country Energy reviews its Bushfire Mitigation Plan annually, as part of last year's review, consideration was given to matters associated with the February 2009 Black Saturday Fires in Victoria coming out of the Royal Commission enquiry. Up to the end of the reporting period, this included consideration of evidence, submissions and the interim findings.

Outcomes of the last annual review included several new initiatives such as:

- Structural changes to provide centralised management of all Country Energy's vegetation clearing and asset inspection activities including the establishment of dedicated roles to ensure ongoing focus on bushfire risk management plans
- A strategic policy change to reduce rural vegetation clearing cycles from a three-year cycle to a consistent two year cycle for all rural vegetation clearing
- A substantial increase in dedicated resources to meet the increased workload associated with this strategy change
- A comprehensive review of the annual bushfire patrol program to strengthen the inspection criteria and ensure the prioritisation of defects in preparation for the bushfire season
- Key modifications to the Asset Management System to increase the visibility of those assets that are within a specified bushfire-prone area. These changes have greatly enhanced the reporting capabilities of the compliance process
- A review of Country Energy's private pole policy targeted at opportunities to limit bushfire risk where private assets are involved. This included a substantial incentive scheme to encourage the undergrounding of 'at risk' overhead private assets
- A review of Country Energy's fire investigation and analysis process to better understand causal data where electrical assets have been involved
- The commencement of nationally accredited Rural Bushfire Service Awareness training for all Country Energy field staff
- Trials of new technology to better understand vegetation profiles, including laser technology and satellite profiling.

A core component of the Bushfire Mitigation Strategy is the Bushfire Risk Mitigation Plan.

The objectives of Country Energy's Bushfire Risk Management Plan are to:

- Ensure public safety
- Establish standards for vegetation management near electricity lines, particularly in bushfire-prone areas
- Reduce electricity supply interruptions related to vegetation
- Minimise the possibility of fire ignition due to electricity lines and associated equipment
- Minimise any other adverse impact of powerlines on vegetation
- Ensure balance between bushfire risk management, vegetation management and the provision of a safe, reliable and adequate supply of electricity of appropriate quality.

8. Bushfire Risk Management

The Bushfire Risk Management Plan covers the following key issues:

- Identification of network assets in bushfire-prone areas capable of initiating bushfires
- Asset maintenance and refurbishment procedures
- Vegetation management strategies and procedures
- Equipment and construction standards as related to bushfire mitigation
- Information relating to the rights and duties of customers who own private powerlines and the dangers of trees coming into contact with those lines
- Asset maintenance procedures for private powerlines and provisions to ensure maintenance standards are enforced by the network operator (Country Energy in this case)
- Management of public enquiries in relation to bushfire mitigation activities such as vegetation clearances, to ensure appropriate investigations and remedial actions are undertaken as required
- Auditing of activities such as asset inspection and vegetation clearances
- Liaison and consultation with other organisations in relation to bushfire risk management issues
- Information for the general public about fire hazards associated with overhead powerlines and vegetation
- Special operating procedures and precautions during times of high fire danger
- Reporting to the NSW Minister for Industry and Investment.

Digital bushfire-prone area mapping data for NSW is available in Country Energy's geographic information system. This information facilitates better operational management during fire ban periods and is also used to target preventative programs, such as the fitting of low voltage spreaders.

8.2 ASSET INSPECTION

Country Energy's asset inspection and assessment process includes:

- Inspection and assessment of overhead poles and structures
- Visual inspection of overhead powerlines, including an assessment of all associated assets such as cross arms, conductors, pole mounted equipment and the above ground components of underground systems
- Treatment of poles
- Earthing integrity checks
- Identification of vegetation encroachment issues.

Private assets are recorded in Country Energy's asset management system and inspected on the same cycle as other assets. At 30 June 2010, the total pole population was 1,387,734 with 78,880 of those being privately owned.

The results of the 2009/2010 asset inspection program are shown Table 8.1.

Table 8.1 – Asset inspection program 2009/2010

Poles inspected	Poles replaced	Poles reinforced
289,478	6,214	939

Country Energy has an average pole replacement/reinforcement rate of 0.52 per cent overall or 2.47 per cent of poles inspected that is consistent with figures for previous years.

8. Bushfire Risk Management

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8.3 ANNUAL BUSHFIRE PATROL

Annual bushfire audits are carried out prior to the normal commencement of the bushfire season using a variety of methods including fixed-wing aircraft, helicopter and ground based patrols. All system defects observed, including those that are not likely to have an associated fire risk, are recorded in the asset management database with an appropriate severity rating indicating the level of risk and the required repair time. 'Urgent risk' defects require rectification within 14 days. 'Risk' defects do not carry an immediate concern for safety or reliability but are scheduled for rectification within six months.

The results of the annual pre-summer bushfire patrols for the 2009/2010 season are shown in Table 8.2.

Table 8.2 – Annual patrol defects identified 2009/2010

Defect category	Urgent risk defects – Rural areas
Number of defects identified in annual patrols	1,955
Defects rectified by 30 June 2010	1,883

The final number completed will be slightly less than that identified at any given point as there is up to 14 days between new defects identified and completion of the task. In the case of table 8.2 the difference of 72 defects open represents the work in progress.

Country Energy completed the urgent risk defects progressively as sightings were identified throughout the patrol period. Individual sightings are generally rectified within 14 days. An Urgent risk report is emailed daily to all relevant managers to ensure they monitor the status of this work and complete tasks within agreed timeframes.

8.4 FIRE REPORTS

All fires attended by Country Energy employees are recorded in a network incidents database, including those in which Country Energy assets have been damaged and/or the alleged cause of ignition involved Country Energy or private line assets.

Table 8.3 – Fire ignitions attended by Country Energy personnel

Bird/animals	13
Storm damage	10
Electrical fault (conductor break, clashing, etc.)	23
Equipment failure or malfunction	50
Pole top fire	16
Vegetation	28
Vehicle impact	6
Customer installation (house, building, switchboard)	93
Bushfire	7
Other	12
Undetermined	16

Total

274

8. Bushfire Risk Management

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Notes:

- Electrical faults include fault current, clashing, conductor break
- Electrical failure or malfunction includes pole, pole top equipment, cross-arm, and fuse failures
- Bushfire – Assets destroyed or damaged by a bushfire where cause does not relate to network assets
- Vegetation includes trees and limbs falling into the clearing space
- Vehicle impact includes farm machinery, high loads and aircraft
- Customer installation includes attendance to disconnect supply for house fires
- Other – A small group of individual fires that does not fit within other categories.

8.5 AUDIT OF ACTIVITIES

Country Energy has full time auditors who conduct field audit of asset inspection activities to ensure consistent understanding of the task requirements and the associated processes are in accordance with organisational expectations. Each asset inspector is audited annually.

Vegetation works conducted by crews are randomly audited on a regular basis by vegetation officers or coordinators to ensure contractual obligations are met. This includes the quality of works undertaken and competency of the workforce. Independent of this a rural annual pre-summer aerial inspection of the lines provides an understanding of the clearances obtained.

Country Energy intends to also implement a specific audit of the vegetation scoping works in the field to ensure there is a consistent understanding of the risk ratings applied to each piece of work identified. An audit of the aerial patrol service providers is conducted to ensure contractual obligations such as coverage of the network and data quality is met.

8.6 COMMUNITY AWARENESS

Specific awareness campaigns that encourage reporting of vegetation in proximity to powerlines by the public are run by Country Energy immediately prior to, and during, the bushfire season.

A general safety media schedule runs throughout the year and targets seasonal activities through print and electronic media, and in specialist industry publications to reach key target groups such as heavy plant operators.

Bushfire-related electrical safety topics included:

- Bushfire and storm safety tips
- Dangers of trees near powerlines
- Electrical safety for emergency services personnel
- Advice on safety when burning off
- Safe operation of farm equipment such as grain augers, harvesters and irrigation systems
- Advice on electrical safety for heavy plant operators.

Country Energy's general education programs include:

- Community consultation through on-site meetings with Country Energy qualified staff to address specific vegetation issues
- Tree planting guidelines
- Advice on vegetation management

8. Bushfire Risk Management

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- Liaison with stakeholders including property owners, government departments, bushfire associations and interested community groups
- Attendance at community meetings
- Presentations at field days and workshops, and
- Bushfire safety information on Country Energy's website.

Refer to section 9 for further information on the public safety awareness management.

9. Public Electrical Safety Awareness Campaign Reports

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9. Public Electrical Safety Awareness Campaign Reports

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9.1 PUBLIC SAFETY – IMPROVED COMMUNITY SAFETY PERFORMANCE

The number of reportable incidents increased slightly from 10 in 2008/2009 to 11 in 2009/2010. Although these statistics show a slight increase in incidents, it is in the light of significant increased agricultural activity due to favourable weather conditions. Country Energy continues to work towards the elimination of all incidents to meet its corporate strategy and to satisfy community expectations.

The graph below shows the declining trend in public safety incidents

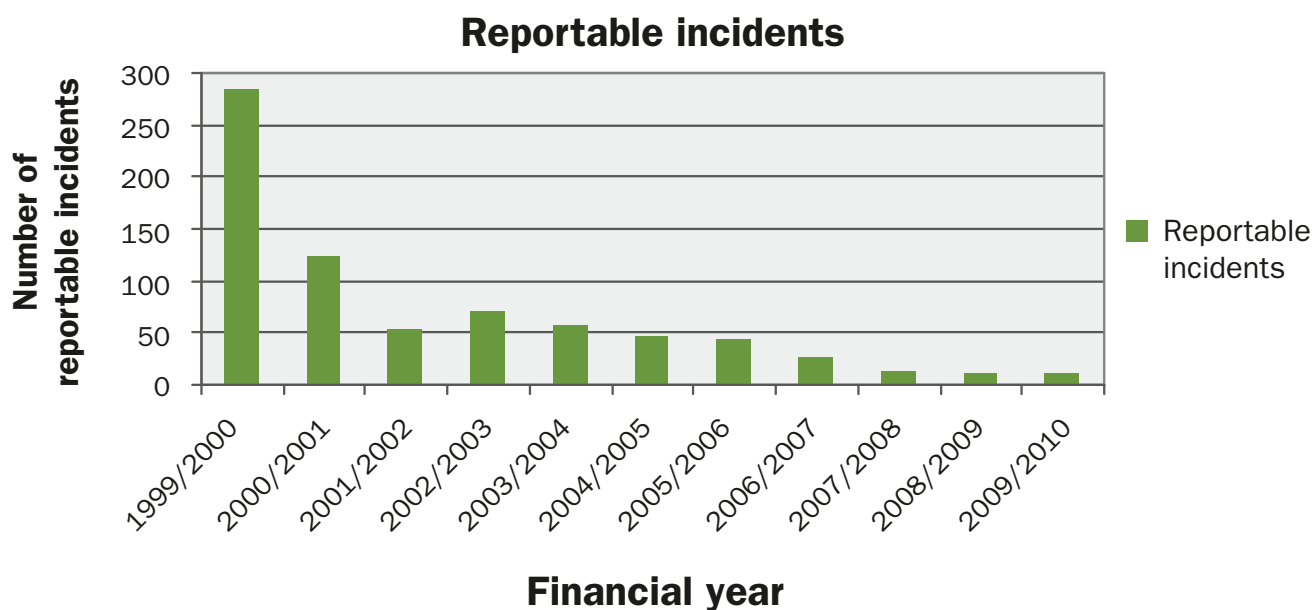


Figure 8 Reported public safety incidents

The strategy of identifying the 'at risk' groups and then educating and assisting those groups to solve their electrical safety issues is a key contributing factor in the decline and maintenance of minimal incidents. These 'at risk' groups include emergency services earthmovers, construction workers, cotton growers, pilots, transport groups, local and state government workers and the like. Regional Safety, Environmental Audit and Coordinators (SEAs) are now responsible for Public Safety within their respective regions.

Key remaining areas of incidents continue to be agriculture, aircraft and construction operations.

9. Public Electrical Safety Awareness Campaign Reports

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Reportable public safety incidents 2009/2010

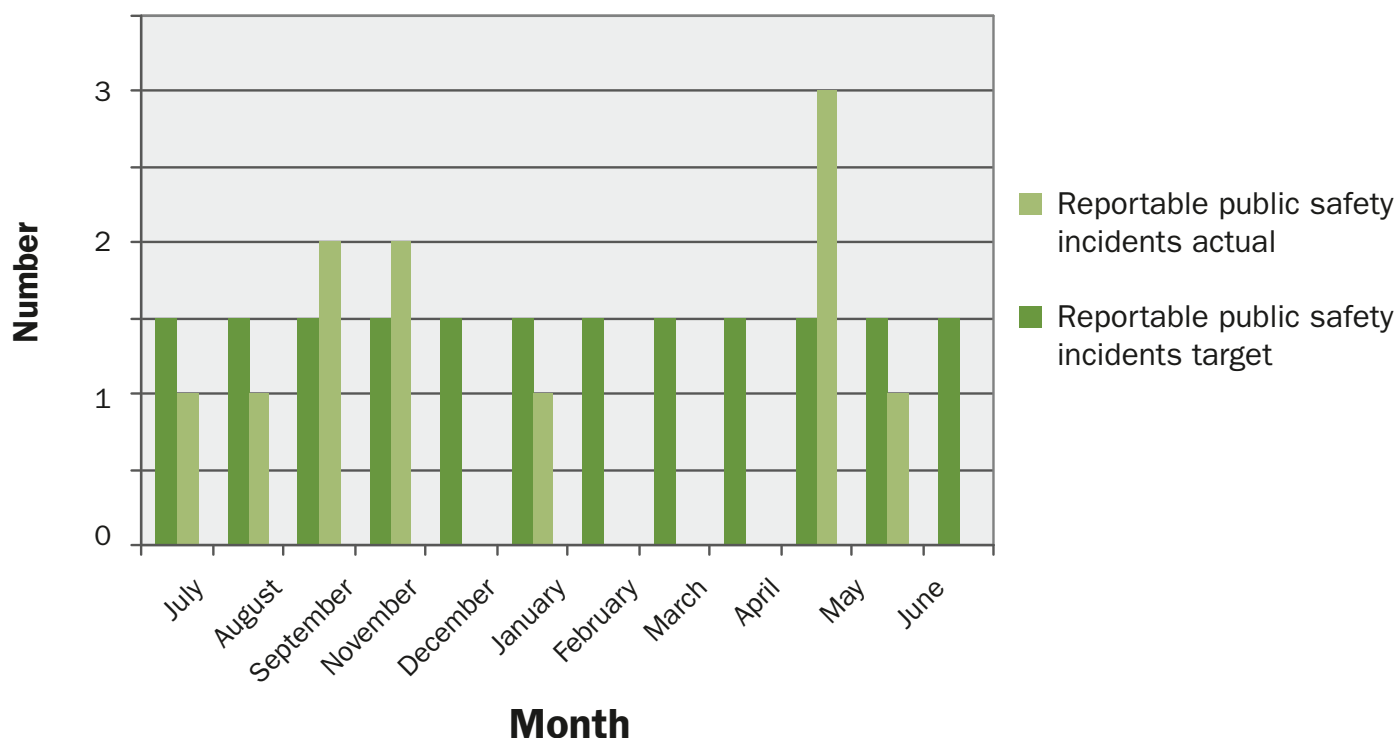


Figure 9 Reportable public safety incidents 2009/10

Five years ago, the Public Safety Team implemented a follow-up survey of a small random sample of the participants to assess training effectiveness. This survey was based on participants remembering the key points delivered during the awareness sessions after a three-month period.

Retention of the key learning outcomes continues to be outstanding, at around 90 per cent over the past year. To help with retention, presenters utilise a powerpoint presentation, practicable demonstrations and a reinforcing DVD exposing the participants to the key points in three different ways.

A recall rate of 15–20 per cent is considered to be normal after three months. This program structure delivers the learning in multiple forms and also delivers an understanding so that the ability to recall is less important in knowing how to respond.

The training by the public safety staff is provided free of charge to the target organisation. After consultation with the organisation, it is presented in a practical and relevant manner targeted directly at the risk group.

9. Public Electrical Safety Awareness Campaign Reports

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During the year the Public Safety Team:

- Provided direct face-to-face electrical safety awareness sessions to more than 1,500 people in 'at risk' groups
- Continued distribution of '3,6,8 clearance' stickers for operators in accordance with the WorkCover Code of Practice – *Working near overhead powerlines*
- Continued to update Public Safety fact sheets on the Country Energy's website giving easy access to important safety information – <http://www.countryenergy.com.au/> - Switch onto Safety: Community, Emergency Services, Agri-Business, Construction, Vegetation Management and Natural Gas Safety
- Updated a static display showing the hidden hazard caused by machinery coming in contact with overhead high voltage powerlines
- Provided safety information packs to individuals and groups involved in line strike incidents
- continued utilisation of a scaled powerline model to demonstrate indications of possible hazards when attending an incident where electricity may be involved
- Attended 11 industry and agricultural field days as well as attending a host of local shows/field days
- Actively promoted the WorkCover Guide to *Work near overhead powerlines* across our region
- Provided electrical hazard awareness training to NSW Rural Fire Service, Forestry Service, earthmoving companies, local government councils
- Participated in 'at risk' group industry working groups to imbed electrical safety into their standard work practices.

Driving the number of incidents lower over the next year will present a challenge with a significant increase in agricultural activity driven by favourable weather patterns.

By the process of direct intervention, the cornerstone of the Country Energy Public Safety Program, Country Energy has achieved a remarkable decline in Public Safety incidents.

The communications strategy necessary to achieve these outcomes is both varied and tailored to the specific 'at risk' group and organisation in order that the intervention is successful and to ensure safe work practices are adopted.

The Country Energy Public Safety Program is reviewed and updated annually and is an accurate picture of the very effective incident reduction strategies that Country Energy has in place.

10. Powerline Crossing of Navigable Waterways

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10. Powerline Crossing of Navigable Waterways

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10.1 EXISTING CROSSING NUMBERS

In consultation with NSW Maritime the total number of crossings of waterways has been revised down from 2,470 sites in the 2008/09 reporting period to 1,357 sites.

These 1,357 sites are part of an ongoing inspection and risk assessment program to establish and record each site as being 'navigable' or 'not navigable' in accordance with the NSW Maritime Crossing of Navigable Waters: Electricity Industry Code.

The initial inspection of these crossings takes into account various aspects of the crossing, including but not limited to:

- The extent to which the waterway is navigable
- The type and size of vessel which can be reasonably expected to navigate the waterway, and
- Other specific details in relation to the site such as the height of the conductors above the waterway and the condition and visibility of crossing signage.

The information gained from the initial crossing inspections is used to carry out a formal risk assessment which classifies the associated risk-rating of the crossing. This classification will then determine how the site will be treated in terms of risk management and will be used in the development and implementation of a prioritised program of crossing relocation, refurbishment and maintenance works.

Table 10.1 – Powerline crossings of navigable waterways summary

	Existing (number)	New (number)	Incidents (number)	Crossings reconstructed (number)	Crossings identified as requiring conversion to submarine crossings (number)
Overhead Crossings	1,302	3	–	–	–
Submarine Crossings	55	–	–	–	–

Note: The existing numbers of overhead and submarine crossings shown in Table 10.1 are based on figures provided by the Network Asset Group using the NSW Maritime GIS overlay.

10.2 PROPOSED PLANNING WORKS

- All new work to be in accordance with the Australian Standard AS 6947-2009 Crossing of waterways by electricity infrastructure, the NSW Maritime Crossings of Navigable Waters: Electricity Industry Code and Country Energy's Procedural Guideline CEPG2013 Crossing of NSW Navigable Waters
- Finalise the inspection and risk assessment of the 1,357 sites mentioned above. This will include an inspection of vegetation, the confirmation of existing data, inspection of supporting structures and hardware, conductor condition, height verification and signage requirements
- Prepare and implement a signage replacement program based on the risk assessment process to install crossing warning signs in accordance with AS6947-2009 Crossing of Waterways by Electricity Infrastructure and the NSW Maritime Crossings of Navigable Waters: Electricity Industry Code
- Update Country Energy's corporate asset database with current inspection details. Note: Country Energy's corporate asset database has been enhanced to record additional site attributes to facilitate effective and efficient management of crossing assets
- Ensure Country Energy employees are aware of the industry standards, codes and reporting protocols that apply with all new and existing crossings

10. Powerline Crossing of Navigable Waterways

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- Ensure external design employees consulting for Country Energy are designing crossings in accordance with the NSW Maritime Code
- Investigate and give consideration to alternative design criteria that will assist in preventing asset corrosion in extreme coastal environments
- Continue to maintain open channels of communication and consultation with the NSW Maritime, relevant government agencies, waterway user and specific interest user groups to ensure foreseeable risks associated with crossings, particularly those to navigation safety, are kept as low as reasonably practicable.

10.3 CROSSINGS INCIDENTS

Country Energy had no recorded incidents for the 2009/2010 reporting period.

APPENDIX 'A'

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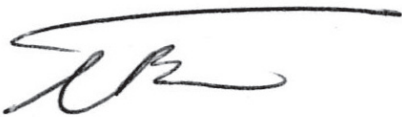
COUNTRY ENERGY ELECTRICITY NETWORK PERFORMANCE REPORT 2009–2010

Declaration by the Managing Director

In submitting this Electricity Network Performance Report (the Report), I declare the Report:

1. Complies with reporting requirements prescribed under the *Electricity Supply (Safety and Network Management) Regulation 2008*, and the 'Distribution Network Service Provider Annual Report Outline' (the Outline), as provided by Industry and Investment NSW
2. Has been appraised by an independent appraiser in accordance with the 'Appraisal Guidelines' contained in the 'Network Operator Report Outline'
3. Includes a copy of the appraisal, signed by the independent appraiser.
4. Has been checked in accordance with recognised quality procedures; and in my opinion, there are reasonable grounds to believe the data, and notes in respect of data contained in this Report, give a true and fair view of the organisation's performance in respect of the matters contained in the 'Network Operator Report Outline'.

SIGNATURE:



Terri Benson
Managing Director

DATE: 15 December 2010